Bank Market Power and Interest Rate Risk *

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November 9, 2026

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Abstract

How does relative market power in funding and lending shape bank interest rate risk exposure and monetary transmission? We document a novel pricing fact: banks with less rate-sensitive funding charge higher fixed-rate loan spread over variable-rate loans, despite stable funding makes fixed-rate lending an attractive earnings hedge. Greater loan market concentration is associated with higher spreads conditional on funding sensitivity and a stronger expansion of fixed-rate credit after monetary tightening. We develop a two-date model in which banks trade higher expected earnings from restricting fixed-rate lending against greater earnings risk due to forgone hedging. A restriction on relative market power reproduces the negative premium–sensitivity slope, while endogenous entry explains why competition in both markets moves together. A calibrated investment extension captures lower premiums and a higher fixed-rate share after an interest rate increase. The extension implies a contraction in aggregate investment. An application shows that greater bank risk-bearing capacity can raise expected earnings and earnings volatility while reducing fixed-rate lending and investment. Banks that can bear more risk become less willing to supply borrower protection, linking bank risk management to monetary transmission.

*I am grateful to Wei Cui for his guidance and support throughout this project. I also thank Adrien Couturier, Thomas Lazarowicz, Marta Morazzoni, Alan Olivi, Mert Onen, Manuel Paillacar-Almonacid, Franck Portier, Lukasz Rachel, Morten Ravn, Guido Spano, Vincent Sterk, Pengfei Wang, Yue Wang, Nianyi Zhang, and seminar participants at UCL for their helpful comments and discussions.

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1 Introduction

The banking turmoil of March 2023, following the rapid rise in U.S. interest rates, renewed concerns about banks' exposure to interest rate risk. Deposit market power dampens the passthrough of higher policy rates to funding costs and hence limits the increase in interest expense. Meanwhile, lending market power impacts interest income response through bank pricing incentives and portfolio composition. How does bank market power on both sides of the balance sheet jointly shape banks' interest rate risk exposure? And how does bank risk management affect monetary transmission?

We link U.S. regulatory data on bank funding and detailed loan-level data, and document a novel pricing fact: banks with less rate-sensitive funding charge a higher fixed-rate premium, the difference between the origination rates on fixed- and variable-rate loans. A one-standard-deviation decrease in interest expense sensitivity is associated with a 5.3-basis-point higher fixed-rate premium. This pattern is puzzling: banks with stable funding costs should value fixed-rate lending as an earnings hedge and therefore charge *lower* premiums. Yet the data show precisely the opposite.

Empirical evidence points to the joint roles of funding and lending market power in this pricing relationship. We show that banks with more rate-insensitive funding operate in more concentrated local lending markets. We find that a one-standard-deviation increase in bank lending market concentration is associated with a 22.5-basis-point higher premium, holding funding sensitivity fixed. Empirical results also confirm bank hedging incentive: higher funding rate-sensitivity is associated with a higher premium, conditional on lending market concentration. These findings suggest that the fixed-rate premium reflects both funding and lending market power.

Relative market power also impacts loan quantities and composition. Following monetary tightening, fixed-rate loan stocks expand while variable-rate stocks contract. The expansion in fixed-rate lending is stronger at banks with greater lending market power, after allowing shock responses to vary with interest expense sensitivity.

To interpret these facts, we build a model with interest-rate uncertainty. The model features risk-averse banks choosing the composition of lending and entrepreneurs selecting fixed- or variable-rate loans. Depositors search and compare offers locally, generating funding-rate passthrough that increases with deposit competition. Banks compete in quantities in a wider fixed-rate lending market, while variable-rate lending is competitive.

The central mechanism is that *relative* market power in funding and lending determines banks' chosen exposure to interest rate risk. Funding market power governs interest expense sensitivity and hence the fixed-rate loan share that hedges earnings. Lending market power can make it profitable to restrict fixed-rate lending and sustain a larger spread over variable-rate loans. Banks thus trade higher expected earnings against forgone hedging and potential

losses following adverse interest rate movements. The key result is that, when lending market power is sufficiently strong relative to funding market power, the model generates a negative premium–sensitivity slope, in line with our stylised fact. Importantly, the model reveals how lending market power impairs risk-sharing between banks and borrowers, with a welfare cost that is *non-monotonic* in interest rate volatility.

We further endogenise bank entry to explain the joint variation in funding sensitivity and lending competition. Larger markets attract more entrants by allowing banks to spread fixed participation costs over greater business volume. Entry increases both deposit-rate sensitivity and competition in fixed-rate lending, with opposing effects on premiums: the former weakens the hedging incentive, while the latter reduces rents from restricting fixed-rate credit. Therefore, in the symmetric entry benchmark, market size can select the degree of competition along the premium–sensitivity relationship derived above.

We embed this mechanism in an equilibrium model of investment and household saving, where fixed-rate protection from interest rate risk supports larger entrepreneurial projects. A higher benchmark rate initially widens deposit margins and attracts bank entry. Greater lending competition makes fixed-rate protection cheaper; at the same time, the entry brings more responsive funding costs, weakening banks' incentive to provide fixed-rate loans as an earnings hedge. Calibrating multiple markets to observed financial conditions and their responses to monetary tightening, we find that the competition effect dominates the hedging effect in the aggregate: borrowers shift towards fixed-rate finance even as higher borrowing costs reduce project sizes. Following a 50-basis-point rate increase, fixed lending rises by 6.03%, while total investment and household deposits fall by 0.67%. Thus, an expansion of fixed-rate credit can coexist with a contraction in total loans and investment, consistent with data. The price of protection governs the composition of borrowing and cushions the investment response, but does not overturn the effect of higher financing costs.

Finally, the same calibration shows that greater bank risk-bearing capacity primarily changes the composition of credit, with little effect on aggregate lending or investment. Banks that can tolerate more earnings risk have less reason to supply fixed-rate loans as a hedge and use their lending power to charge more for protection. Holding bank counts and expected rates fixed, an 80% increase in effective capacity raises the fixed-rate premium by 6.45 basis points and lowers the aggregate fixed share by 5.01 percentage points. Fixed lending falls by 26.01%, but floating lending rises by 6.18%, leaving total lending and investment only 0.02% lower. Borrowers mainly switch contracts, while wage adjustment moderates the change in project scale.

This result identifies a limit to policies that seek to improve borrower risk sharing by increasing banks' capacity to bear risk. Diversification or anticipated public support may raise effective capacity, but banks need not use it to insure borrowers: weaker hedging incentives can instead make fixed-rate protection more expensive and shift interest rate risk

towards entrepreneurs. Through this channel, greater capacity delivers little aggregate credit stimulus and can reduce borrower protection. Its effectiveness therefore depends on banks' incentives to supply insurance, shaped by relative funding and lending market power, as well as their ability to absorb risk.

Relationship to literature. Our paper contributes to three strands of literature: theories of bank interest rate risk, bank market power, and risk sharing through loan contracts.

Theories of bank interest rate risk. Hellwig (1994) studies the efficient allocation of interest rate risk under liquidity provision, showing that eliminating exposure can sacrifice reinvestment opportunities. Drechsler et al. (2021) show that fixed-rate assets can stabilise earnings when deposit franchises insulate funding costs from market rates. Di Tella and Kurlat (2021) explain exposure through a dynamic hedging motive: higher rates raise deposit-intermediation profits, making banks willing to accept contemporaneous capital losses. Varraso (2025) links endogenous maturity choices to bank net worth and financial constraints in a competitive intermediation model.

We identify a complementary mechanism through loan-market pricing power. Banks trade the earnings hedge from fixed-rate lending against the rents preserved by restricting its supply. Relative funding and lending market power governs this tradeoff, linking banks' chosen exposure to the price of borrower protection. Our focus on earnings exposure differs from the market-value sensitivity of bank franchises emphasised by DeMarzo et al. (2024).

Bank market power. A large and growing literature studies bank market power on both sides of the balance sheet. Funding-market power shapes monetary transmission (Drechsler et al., 2017) and financial fragility (Egan et al., 2017), while lending-market power affects mortgage-rate pass-through and refinancing (Scharfstein & Sunderam, 2016), as well as corporate borrowing rates and aggregate activity (Bocola et al., 2026). Wang et al. (2022) jointly model deposit and loan competition to study monetary transmission. Hachem and Song (2021) show how interbank market power shapes liquidity choices and can make tighter liquidity regulation generate a credit boom.

We contribute by examining how funding and lending market power jointly shape banks' interest rate risk management, motivated by our evidence that lending concentration is associated with higher fixed-rate premia conditional on funding sensitivity. We combine deposit market power arising from search frictions, as in Choi and Rocheteau (2023) and Head et al. (2025), with Cournot competition in lending. Endogenous bank entry then provides a common source of variation in competition across the two markets. Related evidence on the importance of entry comes from Begenau and Stafford (2025), who show that branch expansion shapes deposit growth during monetary tightening.

Loan contracts and risk sharing. Fixed- and variable-rate loans allocate interest rate risk between banks and borrowers (Santomero, 1983). Vickery (2008) finds that financially

constrained small firms are more likely to borrow at fixed rates, consistent with insurance demand. Kirti (2020) shows that banks with more rate-sensitive liabilities supply more floating-rate credit and price it more favourably relative to fixed-rate credit. Our evidence points to the opposite pricing relationship. Differences in lending competition may reconcile these findings: Kirti's pricing evidence uses residential mortgage quotes, whereas we study small business loans. Our model reproduces both relationships by varying relative funding and lending market power, highlighting its role in allocating interest rate risk.

Liu and Elenev (2025) connect mortgage fixation to financial stability and risk sharing; Abad et al. (2026) examine how loan fixation and bank solvency shape monetary transmission; and Bretscher et al. (2018) link interest rate uncertainty to corporate hedging and investment.

We extend these accounts of borrower insurance demand and banks' funding incentives by jointly determining fixed-rate loan pricing, borrower contract choice, and investment in our quantitative model. Greater bank risk-bearing capacity weakens the incentive to hedge earnings through fixed-rate lending. Banks restrict fixed credit and raise premiums, accepting greater earnings risk for higher expected earnings. More expensive protection reduces investment. This mechanism connects banks' capacity to bear risk to the supply of borrower protection and real activity.

2 Stylised Facts

Our empirical analysis combines loan-level records from the U.S. Small Business Administration's (SBA) 7(a) programme, bank regulatory reports (Call Report), and the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits. We consolidate affiliated institutions at the ultimate holding-company level, allowing for funding and asset allocation to be coordinated across subsidiaries through internal capital markets (Campello, 2002). Throughout the paper, "bank" refers to this consolidated entity.

The SBA records provide detailed loan-level information, including each loan's rate type and interest rate at origination. They offer a detailed window into small-business lending, a quantitatively important segment of bank credit: loans below $1 million accounted for 22% of domestic banks' commercial and industrial loan origination.[1] Small firms' dependence on bank finance and limited use of derivatives make their loan terms particularly consequential for managing interest rate exposure (Vickery, 2008). These features make small-business lending a useful setting for studying lending market power.

Call Report provides bank-level funding information, including interest expense, whose

[1]By value in May 2017. Calculated from Table 2 of the Federal Reserve's *May 1–5, 2017 Survey of Terms of Business Lending*. The relevant category covers loans of $10,000 to below $1 million; loan size serves as a proxy for small-business lending.

sensitivity to interest rates we use to assess funding market power following Drechsler et al. (2021). Call Reports also provide information on the repricing profile of outstanding loans. For the quantity analysis, we follow Kirti (2020) and proxy variable-rate loan stocks by loans maturing or repricing within one year, and fixed-rate stocks by the remaining loans. From the Summary of Deposits, we construct a county-level branch-count Herfindahl–Hirschman index (HHI) by summing banks' squared shares of local branches. We average these county indices equally across the counties in each bank's branch network to measure the local banking concentration it faces. Given the local nature of SBA lending,[2] we use this concentration measure as a proxy for lending market power, once conditional on funding sensitivity.[3] Appendix A provides detailed descriptions of the data and variable construction.

We document two stylised facts about the pricing and quantity of fixed-rate credit. First, banks with less rate-sensitive funding charge higher fixed-rate premiums. In the loan-level regressions, greater local banking concentration is associated with a larger fixed–rate premium, conditional on funding sensitivity and observed loan and bank characteristics. Second, fixed-rate loan stocks expand following monetary tightening, while variable-rate loan stocks initially contract. The expansion of fixed-rate lending is larger at banks operating in more concentrated markets. Together, these patterns point to the relative funding and lending market power, as a determinant of the pricing and provision of fixed-rate credit.

2.1 Fixed-rate premium and relative market power

Banks with less rate-sensitive funding charge higher fixed-rate premia. Figure 1(a) shows that a one-standard-deviation increase in interest expense sensitivity, β_b^{exp}, is associated with a 5.3-basis-point lower premium. This relationship runs against the prediction from earnings hedging alone. When funding costs respond little to interest rates, fixed-rate lending helps stabilise bank earnings, giving banks an incentive to offer these loans at lower premia. The observed negative slope therefore suggests that other pricing incentives vary systematically with funding sensitivity.

One possible explanation lies in differences in lending competition. Figure 2(a) maps county-level concentration measured using banks' shares of SBA 7(a) loan dollars and reveals substantial geographic variation. Some local markets have lending distributed across many banks, while others concentrate lending among relatively few providers. This variation points to differences in the competitive conditions under which banks price small-business loans, potentially affecting their willingness to supply fixed-rate protection.

At the bank level, Figure 1(b) connects lending concentration to funding sensitivity.

[2]Di and Pattison (2020) report that 71% of loans in their matched 2017 SBA sample went to borrowers located within ten miles of a branch of their lender.

[3]Funding sensitivity therefore reflects broader funding conditions, while branch-network concentration captures local competition for borrowers. Appendix A develops this interpretation.

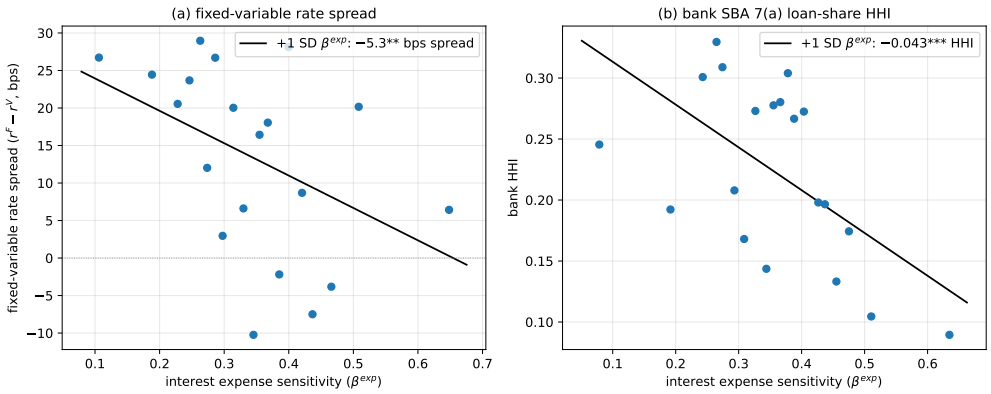


Figure 1: Bank funding sensitivity, fixed-rate premia and lending concentration

Notes: Panel (a) plots the fixed–variable origination-rate differential against bank interest expense sensitivity. Panel (b) plots bank-level SBA 7(a) loan-share HHI against the same sensitivity measure. Dots represent binned observations, and solid lines show fitted relationships. Annotations report the change in each outcome associated with a one-standard-deviation increase in interest expense sensitivity. The loan-share HHI in Panel (b) differs from the lagged branch-count HHI used in the pricing regressions. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, and the Small Business Administration (SBA) 7(a) loan programme, 2010Q1–2019Q4.

Banks with less rate-sensitive funding also operate in more concentrated SBA lending markets: a one-standard-deviation increase in interest expense sensitivity is associated with a 0.043 decline in bank-level SBA loan-share HHI. Stable funding thus tends to coincide with greater lending-market power. Banks with lending power may prefer to restrict fixed-rate lending and preserve higher premia, sacrificing some earnings stability to increase expected earnings. This pricing incentive could counteract the hedging incentive associated with stable funding and help explain the negative premium–sensitivity slope in Figure 1(a).

To assess the role of relative market power, we relate loan-level origination rates to banks' funding sensitivity and one-year-lagged branch-count HHI.[4] County-level branch-count concentration is mapped in Figure 7 in the Empirical Appendix. Figure 2(b) shows that branch-count concentration is positively associated with SBA lending concentration, supporting its relevance for the local lending environment. Using branch counts also mitigates a specific simultaneity concern: contemporaneous SBA lending shares reflect banks' pricing and supply decisions together with borrower demand, so loan-share HHI may be jointly determined with the fixed–variable rate differential. Lagged branch-count HHI captures established banking presence without incorporating current lending volumes. We therefore use it as a proxy for lending-market power conditional on funding sensitivity and the other controls. Since branch networks may still be endogenous, we interpret the estimates as conditional associations rather than causal effects.

[4]The pricing sample starts in 2010Q1 with SBA rate-type reporting and ends in 2019Q4 to focus on relatively stable policy-rate uncertainty (Bauer et al., 2022).

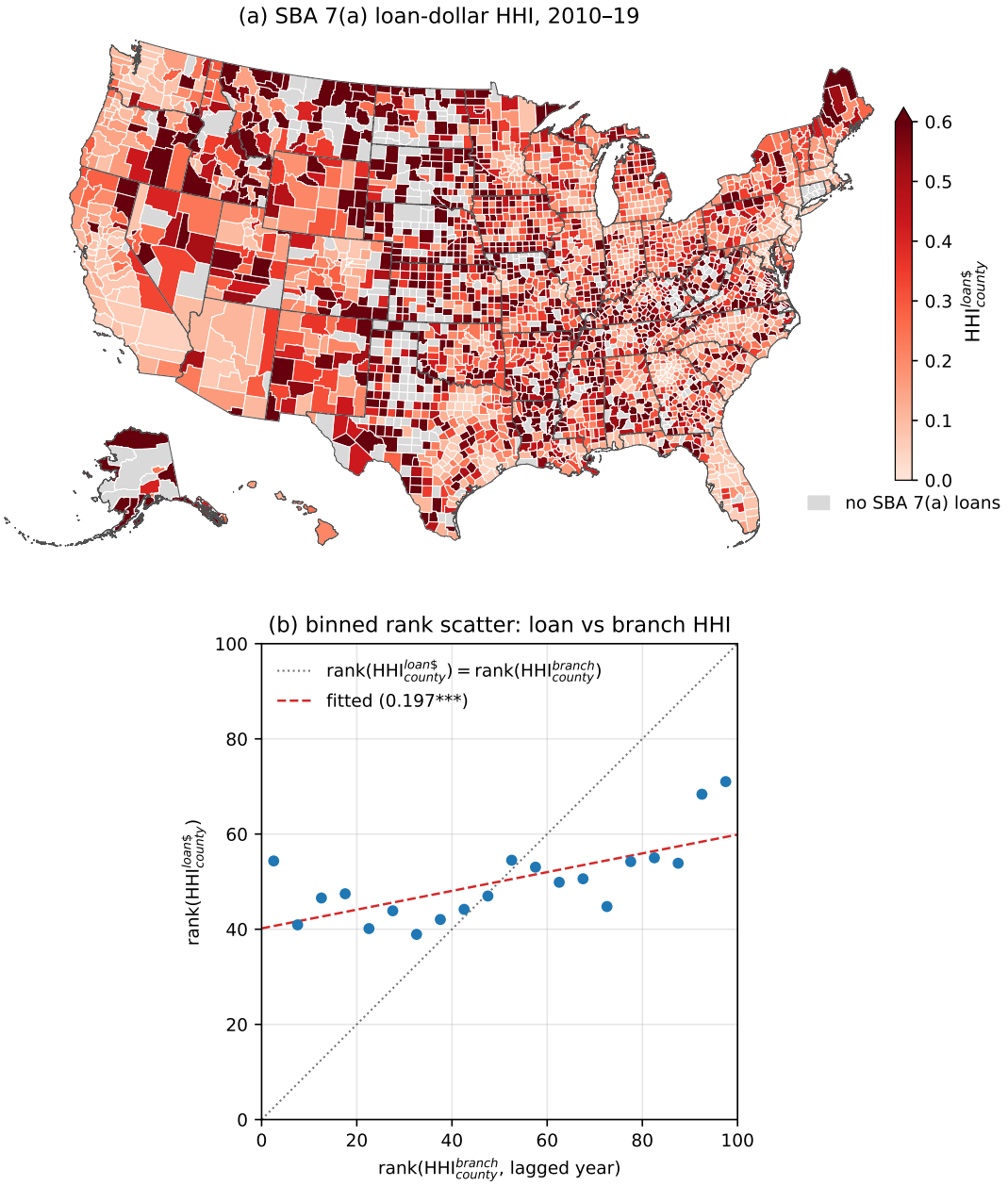


Figure 2: County-level HHI: SBA 7(a) loan share and branch count

Notes: Panel (a) maps county-level concentration measured using banks' shares of SBA 7(a) loan dollars. Grey counties have no SBA 7(a) loans. Panel (b) plots binned county-level ranks of loan-share HHI against ranks of one-year-lagged branch-count HHI. The dashed line shows the fitted relationship, and the dotted line marks equal ranks. Sources: SBA 7(a) loan programme and FDIC Summary of Deposits, 2010–2019.

For loan i of rate type $k \in \{F, V\}$, issued by bank b in quarter t, we estimate

$$\begin{aligned} r_{i,k,b,t} = \delta F_i + \alpha_\beta \beta_b^{\exp} + \alpha_H \mathrm{HHI}_{b,t-4} \\ + d_\beta F_i \beta_b^{\exp} + d_H F_i \mathrm{HHI}_{b,t-4} \\ + X_i'\gamma + F_i X_i'\gamma_F + W_{b,t}'\psi + F_i W_{b,t}'\psi_F \\ + \eta_{\text{state},t} + \eta_{\text{industry}} + \varepsilon_i. \end{aligned}$$

Here, $F_i = \mathbf{1}\{k = F\}$ indicates a fixed-rate loan. Loan characteristics X_i include the guaranteed share, jobs supported and maturity, while bank characteristics $W_{b,t}$ include log assets and the equity-to-assets ratio. Their interactions with F_i allow these characteristics to have different associations with fixed- and variable-rate pricing. We include state-by-quarter and borrower-industry fixed effects and cluster standard errors by bank. The interaction coefficients d_β and d_H capture the relationships of funding sensitivity and concentration with the fixed–variable rate differential; the coefficients α_β and α_H capture their relationships with variable-rate pricing.

Table 1: Fixed- and variable-rate loan interest rates, interest expense sensitivity and one-year-lagged branch-count HHI

Dependent variable: interest rate at origination $r_{i,k,b,t}$ (%)	(1)	(2)	(3)	(4)	(5)
	OLS	quarter, industry FE	state×qtr, industry FE	loan-level controls	bank-level controls
Interest expense sensitivity (β_b^{exp})	-0.731	-0.252	-0.503*	-0.462	-0.193
	(0.506)	(0.292)	(0.285)	(0.291)	(0.303)
HHI, lagged one year ($\mathrm{HHI}_{b,t-4}$)	-0.006	-0.044	-0.733	-0.976	0.154
	(1.394)	(0.803)	(0.861)	(0.910)	(0.952)
$F_i\times$ interest expense sensitivity	-0.197	-0.097	0.048	0.656	1.106
	(0.933)	(0.811)	(0.762)	(0.808)	(1.013)
$F_i\times$ HHI (lagged)	1.839	2.452	2.755	3.267*	3.866***
	(2.735)	(1.878)	(1.715)	(1.843)	(1.451)
Loan-level characteristics (X_i, F_iX_i)	No	No	No	Yes	Yes
Bank-level characteristics ($W_{b,t}$, $F_iW_{b,t}$)	No	No	No	No	Yes
Industry FE	No	Yes	Yes	Yes	Yes
State × quarter FE	No	No	Yes	Yes	Yes
Observations	57,018	57,018	57,018	57,018	57,018
Bank-year cells	835	835	835	835	835
Banks	307	307	307	307	307
R^2	0.011	0.382	0.432	0.463	0.481

Notes: standard errors clustered by banks in parentheses; *, **, ***: $p < 0.10$, 0.05, 0.01. Sample: SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks who had a branch network in the Summary of Deposits one year before the approval-year snapshot, restricted to bank × calendar-year cells with at least 2 fixed- and 2 variable-rate loans and fewer than 20% of their variable-rate loans at the statutory cap. The interest rate at origination, bank interest expense sensitivity, HHI and loan- and bank-level characteristics are winsorized at the 1st/99th percentiles within rate type. Interest expense sensitivity, $\beta_b^{\exp}$, and bank-level HHI, $\mathrm{HHI}_{b,t}$, are constructed as described in Appendices A.2 and A.3, respectively.

Table 1 reports estimates for 57,018 loans across 835 bank-year observations from 307

banks. The interaction between fixed-rate status and HHI is positive in all five specifications. In the preferred specification, Column (5), its coefficient is 3.866 with a standard error of 1.451, significant at the 1% level. A one-standard-deviation increase in bank lending-market concentration is associated with a 22.5-basis-point higher fixed-rate premium, holding funding sensitivity and the included loan and bank characteristics constant. This finding is consistent with lending-market power supporting higher premia for fixed-rate protection.

The preferred specification also yields a positive interaction between fixed-rate status and funding sensitivity: the coefficient is 1.106 with a standard error of 1.013. Its direction is consistent with the earnings-hedging mechanism. When funding costs are more rate-sensitive, an increase in those costs cannot be matched by repricing fixed-rate loan income, making fixed-rate lending less attractive to banks and increasing the premium they require. The estimate is not statistically distinguishable from zero, however, so the evidence for this conditional funding relationship remains suggestive.

For variable-rate loans, neither funding sensitivity nor concentration has a statistically detectable relationship with pricing in the preferred specification. The corresponding coefficients are -0.193 with a standard error of 0.303 for funding sensitivity and 0.154 with a standard error of 0.952 for HHI. Thus, the clearest pricing association is between concentration and the premium on fixed-rate loans relative to variable-rate loans.

Taken together, these facts highlight the relevance of relative market power in funding and lending for fixed-rate pricing. Stable funding creates an incentive to supply fixed-rate loans as an earnings hedge, while lending-market power creates an incentive to preserve premia by restricting their supply. The concentration result and the direction of the conditional funding-sensitivity estimate are consistent with these opposing influences. They motivate a theoretical model in which banks weigh expected earnings against earnings stability, determining both the fixed-rate premium and the allocation of lending between fixed- and variable-rate contracts.

2.2 Lending composition and relative market power

Having established the relationship between relative market power in funding and lending and the fixed-rate premium, we now turn to loan quantities. We examine how monetary tightening changes the stocks of fixed- and variable-rate loans, and whether the fixed-rate response differs across banks with different lending market power, conditional on their funding sensitivity. We use the monetary policy shocks of Bauer and Swanson (2023), scaled to a contractionary shock of 25 basis points. We use 2002Q2–2024Q4 to exploit more monetary policy variation and improve estimation precision.

Let $l^k_{b,t}$ denote the outstanding stock of loans of rate type $k \in \{F, V\}$ held by bank b in quarter t. We estimate cumulative responses separately by rate type using the following

local projection specification:

$$\log l^k_{b,t+h} - \log l^k_{b,t-1} = \phi^k_h \epsilon^{MP}_t + W'_{b,t-1}\Gamma^k_h + Z'_{b,t-1}\Lambda^k_h + \alpha^k_b + \varepsilon^k_{b,t,h},$$

Here, h is the horizon in quarters and ϵ^{MP}_t is the monetary policy shock. The coefficient ϕ^k_h measures the cumulative response of loan stocks from the quarter preceding the shock to horizon h. The vector $W_{b,t-1}$ contains lagged bank characteristics, including asset size, leverage, interest expense sensitivity and HHI. The vector $Z_{b,t-1}$ contains lagged aggregate controls, including the one-year Treasury yield, output growth and the yield-curve slope. We also include bank fixed effects, α^k_b.

Panel (a) of Figure 3 shows that fixed- and variable-rate loan stocks respond differently to monetary tightening. Fixed-rate stocks increase over the first year, reaching approximately 2.6% above their pre-shock level after four quarters, while variable-rate stocks fall by approximately 5.4%. The fixed-rate increase subsequently fades, whereas variable-rate stocks remain below their pre-shock level throughout the remainder of the two-year horizon. Monetary tightening therefore shifts the composition of outstanding lending towards fixed-rate credit.

We next examine whether lending market power is associated with this adjustment. We allow the fixed-rate response to differ across quintiles of bank-level HHI, our proxy for lending market power. We also interact the monetary policy shock with interest expense sensitivity, allowing banks' responses to vary with their funding characteristics. This specification compares the responses across lending-market concentration groups after accounting for differential responses associated with funding sensitivity:

$$\begin{aligned}\log l^F_{b,t+h} - \log l^F_{b,t-1} = &\sum_{q=1}^{5} \phi^F_{h,q} \epsilon^{MP}_t \times 1[\text{quintile}(b) = q] + \theta_h \epsilon^{MP}_t \beta^{\text{exp}}_b \\ &+ W'_{b,t-1}\Gamma^F_h + Z'_{b,t-1}\Lambda^F_h + \alpha^F_b + \varepsilon^F_{b,t,h},\end{aligned}$$

Here, $1[\text{quintile}(b) = q]$ indicates bank b's HHI quintile. The quintile-specific coefficients, $\phi^F_{h,q}$, capture the fixed-rate response within each HHI group, while θ_h captures how the response varies with interest expense sensitivity. The remaining controls and bank fixed effects follow the first specification.

Panel (b) of Figure 3 shows that banks with greater lending market power exhibit the stronger increase in fixed-rate loan stocks. After four quarters, the estimated increase is approximately 4.2% at high-lending-power banks, compared with 0.8% at low-lending-power banks. The high-power response remains larger over subsequent quarters. Conditional on funding sensitivity, the expansion of fixed-rate lending following monetary tightening is therefore more pronounced among banks with greater lending market power.

Together, the pricing and quantity evidence connects relative market power to both the

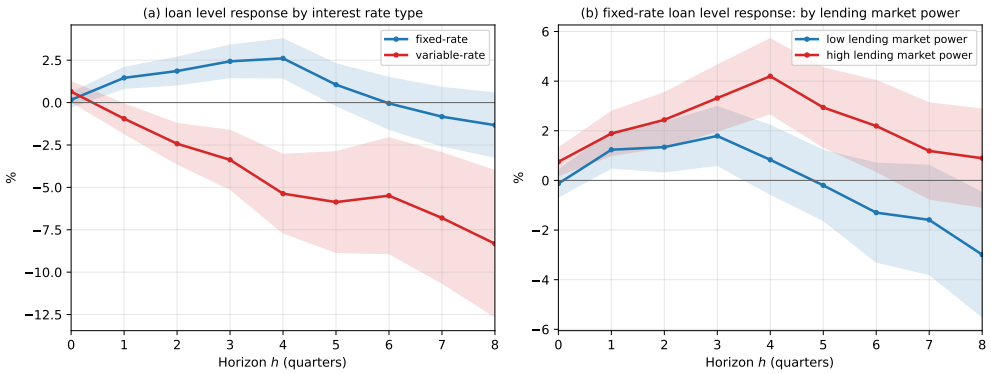


Figure 3: Fixed- and variable-rate lending following +25bps monetary tightening

Notes: Local-projection responses to a contractionary monetary policy shock from Bauer and Swanson (2023), scaled to +25 basis points. Panel (a) compares fixed- and variable-rate loan stocks; Panel (b) compares fixed-rate loan stocks at banks with low and high lending market power, measured by bank-level HHI. Responses are expressed as 100 times cumulative log changes and can be read approximately as percentage changes in loan stocks. Horizons are in quarters, and shaded bands are 68% confidence intervals. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, 2002Q2–2024Q4.

pricing of fixed-rate premiums and banks' adjustment of their loan portfolios. Section 3 examines how banks' tradeoff between lending rents and earnings hedging determines fixed-rate premiums and contract choice. Section 4 takes the pricing mechanism to an equilibrium model. By allowing borrowing and investment adjustment, we assess how this mechanism shapes both the composition of lending and the response of overall credit provision to higher interest rates.

3 Model

In this two-date model, stable deposit costs give banks a reason to offer fixed-rate loans at a low premium: when funding costs move little with rates, fixed-rate income helps stabilise earnings. Lending power creates an opposing incentive. To attract more fixed-rate borrowers, a bank must lower the premium on its other fixed-rate loans. The portfolio that best hedges earnings can therefore differ from the one that maximises expected profits. We first hold bank counts fixed, then use entry to explain why deposit and lending competition move together.

3.1 Setup

Loans originate at $t = 0$ and pay at $t = 1$. The only aggregate uncertainty is the net benchmark rate,

$$r_1(z) = r_0 + \sigma z, \qquad \Pr(z = 1) = \Pr(z = -1) = \tfrac{1}{2}, \qquad \sigma > 0. \tag{1}$$

Write $x(z)$ for the outcome in state z. Expectations average over the two states: $\mathbb{E}[x(z)] = [x(-1) + x(1)]/2$. Thus $\mathbb{E}[r_1(z)] = r_0$ and $\text{Var}(r_1(z)) = \sigma^2$. All contracts are chosen before z is realised. The parameters and distributions below are common knowledge.

Entrepreneurs. A unit mass of entrepreneurs each borrows one unit to finance a project with deterministic output, fixing total lending at one. Output net of principal repayment, y, ensures participation and repayment in both states. Borrowers choose between fixed and variable rates. They receive no deposit income and have no other hedging instruments.

Entrepreneurs have mean–variance preferences over terminal wealth and heterogeneous risk aversion, $\tilde{\theta} \sim U[0, 2\theta]$, with $\theta > 0$. Each knows its type. A fixed-rate loan charges $r^F = r_0 + s$, where s is the premium set at origination; a variable-rate loan charges $r^V(z) = r_1(z)$. The pricing asymmetry reflects bank market power in fixed-rate loan market relative to variable-rate loan market.[5] Their certainty equivalents are

$$CE^F = y - r_0 - s, \qquad CE^V(\tilde{\theta}) = y - r_0 - \frac{\tilde{\theta}}{2}\sigma^2.$$

Borrowers choose a fixed rate when $CE^F > CE^V$, or equivalently, $\tilde{\theta} \geq \theta^* \equiv 2s/\sigma^2$. Rate-type sorting based on risk aversion resembles Campbell and Cocco (2003). For an interior allocation, fixed-rate demand and its inverse are

$$L^F = 1 - \frac{s}{\theta\sigma^2}, \qquad s(L^F, \sigma) = \theta\sigma^2(1 - L^F), \qquad \theta^* = 2\theta(1 - L^F). \tag{2}$$

Demand is one for $s \leq 0$ and zero for $s \geq \theta\sigma^2$; equilibrium will be interior. Expanding fixed-rate lending requires attracting less risk-averse borrowers, who will buy protection only at a lower premium. Throughout, $s = r^F - \mathbb{E}[r^V(z)]$ denotes the expected fixed-minus-variable premium.

Deposit and lending markets. There are $M \geq 1$ local deposit markets and a common lending market. With integer bank counts, each local market has n banks and the total is $N = Mn$. We allow $n \geq 1$ and $N = Mn$ to vary continuously as effective competition indices: n determines access to competing deposit offers, while $1/N$ measures an individual lender's influence on the fixed-rate premium. Integer values admit the symmetric bank-count interpretation; non-integer values use its continuous extension. Depositors compare offers locally, whereas borrowers access the common lending market. We initially treat n and M as primitives. Section 4 allows heterogeneous local markets and segments lending as well as deposits.

[5]The protection offered by fixed-rate financing has few accessible substitutes: Vickery (2008) documents financially constrained small firms disproportionately choose fixed-rate loans and seldom use derivatives to hedge interest rate risk. Public bond market is often costly to access for smaller firms (Faulkender, 2005).

Total deposits are one, divided equally across banks in the integer benchmark, each with predetermined loan capacity $d_i = 1/N$. Depositors are risk-neutral and have an outside savings return $r_a > 0$. This exogenous reservation rate is known at origination and fixed across interest-rate states, competition indices, and market sizes. Its provider and pricing are outside the model. Households cannot directly access the wholesale benchmark $r_1(z)$. We assume $r_0 > r_a$: depositors compare expected returns, so the benchmark need not exceed r_a in every state. We focus on parameter configurations in which banks do not voluntarily increase deposit pass-through to hedge earnings. Appendix C.3 gives sufficient conditions and their compatibility with a negative premium–beta relationship.

Deposit search and pricing. Each depositor has an offer from its incumbent and compares it with a rival's offer if one arrives before the decision deadline. Measure search time in units that give the deadline an exponential arrival rate of one. Each of the $n-1$ rivals makes contact independently at rate $\zeta > 0$ in these units. Thus ζ measures the contact rate per rival relative to the deadline rate. With a continuum of accounts, the fraction receiving a competing offer is

$$p(n;\zeta) = \frac{\zeta(n-1)}{1+\zeta(n-1)}. \tag{3}$$

Throughout this section we assume equal contact and deadline rates, $\zeta = 1$, and write $p(n) = p(n;1) = 1 - 1/n$. Search is exogenous and independent of z and borrower types. The aggregate contact rate extends to real $n \geq 1$; the constant market-breadth threshold below uses $\zeta = 1$.

Eligible banks know whether a depositor has a competing offer and quote personalised indexed contracts $r_a + \alpha(r_1(z) - r_a)$, with $\alpha \geq 0$. Here α is account-level passthrough: $\alpha = 0$ pays r_a, while $\alpha = 1$ pays $r_1(z)$. Since $r_0 > r_a$, risk-neutral depositors prefer larger α and remain with the incumbent when expected returns tie, including a tie with outside savings. Quotation is costless and separate fees are excluded. Banks can replace a lost deposit by borrowing at $r_1(z)$, and invest a newly acquired surplus deposit at $r_1(z)$. This wholesale market accommodates deposit movements against committed loan books; households cannot access it directly.

Competition bids contested accounts up to $r_1(z)$, eliminating their funding advantage relative to wholesale borrowing. Switching at this rate would leave the incumbent's funding cost unchanged, while the receiving bank could invest the extra deposit at the same rate. In the maintained regime, uncontested accounts receive r_a. The tie-breaking rule keeps deposits with their incumbents: competition changes rates even without realised deposit flows. Appendix C.2 verifies optimal bids, allowing for their hedging benefits; $r_1(z)$ is an equilibrium

bidding limit, not an imposed cap. Equilibrium average funding costs and passthrough are

$$r^d(z) = (1-\beta^d)r_a + \beta^d r_1(z), \qquad \beta^d \equiv \frac{\partial r^d(z)}{\partial r_1(z)} = p(n) = 1 - \frac{1}{n}.$$

More local competitors expose more accounts to rival offers, making funding costs more responsive to rates. As $n \to \infty$, banks pass through rate changes one-for-one, i.e. $\beta^d \to 1$. In contrast, when there is a monopoly bank, i.e. $n = 1$, the bank does not pass through any rate changes, i.e. $\beta^d = 0$. The exogenous r_a sets the funding-cost floor without directly changing pass-through; a rate-sensitive outside return would change this conclusion. The construction uses personalised bids and observable comparison sets, rather than a Burdett–Judd posted-rate equilibrium.

Lending and bank preferences. In the integer benchmark, bank i chooses its fixed-rate portfolio share $l_i^F \in [0,1]$. Its fixed quantity and aggregate fixed lending are

$$q_i^F = d_i l_i^F = \frac{l_i^F}{N}, \qquad L^F = \sum_{i=1}^{N} q_i^F = \frac{1}{N}\sum_{i=1}^{N} l_i^F.$$

The remainder of its book is lent at the variable rate. Banks commit to fixed quantities and compete in Cournot fashion, internalising $\partial s/\partial l_i^F = -\theta\sigma^2/N$. The continuous extension retains this price-impact coefficient in the symmetric pricing conditions. All borrowers face the same fixed premium, while banks take $r^V(z) = r_1(z)$ as given. Borrower types can be allocated across banks to fill their capacities.

With retained deposits and bank-specific average pass-through β_i^d, net interest earnings per unit of assets are $\pi_i(z) = l_i^F r^F + (1-l_i^F)r^V(z) - r_i^d(z)$. Substituting the contract rates gives

$$\pi_i(z) = (1-\beta_i^d)(r_0 - r_a) + l_i^F s(L^F, \sigma) + (1-\beta_i^d - l_i^F)\sigma z. \tag{4}$$

Principal repayments cancel. The sensitivity gap is

$$G_i \equiv \frac{\partial \pi_i(z)}{\partial r_1(z)} = (1-l_i^F) - \beta_i^d,$$

so earnings variance is $\sigma^2 G_i^2$. The gap captures the difference between interest income and interest expense sensitivities. A positive gap means that higher rates raise bank earnings; a negative gap means that they lower earnings. Banks maximise

$$\max_{l_i^F \in [0,1]} \left\{ \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \right\}.$$

Throughout, γ applies to earnings per unit of assets (Hart & Jaffee, 1974). Credit losses, cap-

ital requirements, prepayment, separate origination fees, and additional hedging instruments are excluded.

Timing. Comparison sets are realised and observed by eligible banks before the rate shock. Banks then choose fixed-loan quantities, anticipating the deposit continuation. The fixed premium clears the loan market and entrepreneurs select contracts. Eligible banks subsequently quote indexed deposit contracts, depositors choose their accounts, and wholesale funding accommodates any movement. All contracts settle after z is realised. In the selected continuation all deposits are retained and wholesale positions are zero.

3.2 Equilibrium

For given integer counts n, N, banks choose portfolios taking rivals' quantities as given and anticipating optimal deposit bids after every feasible portfolio profile. Borrowers and depositors choose their preferred available contracts, and loan and funding balances clear. We select the deposit retention continuation above and focus on symmetric lending, with $l_i^F = L^F$ and $G = 1 - \beta^d - L^F$. For non-integer competition indices, the corresponding symmetric conditions define the continuous extension.

Lemma 1 (Fixed-rate pricing in symmetric Cournot equilibrium)**.** *Fix n and N and suppose the deposit-retention continuation yields $\beta^d = 1 - 1/n$ at equilibrium. In the symmetric Cournot equilibrium, the fixed-rate premium and the common fixed-loan share $l_i^F = L^F$ satisfy*

$$s = \underbrace{\frac{\theta\sigma^2 L^F}{N}}_{\text{Cournot term}} - \underbrace{\gamma\sigma^2 G}_{\text{marginal hedging benefit}}. \tag{5}$$

Proof. See Appendix C.2. □

When funding costs respond less to rates than loan income ($G > 0$), replacing variable-rate loans with fixed-rate loans stabilises earnings. The bank will accept a lower premium for this hedge. Yet attracting another borrower requires a lower premium on all the fixed loans it originates. The Cournot term measures the premium revenue forgone on its other loans, which the bank weighs against the additional loan's income and hedging benefit. Banks choose fixed quantities at origination, when the common premium clears the market. Existing fixed loans retain their contracted rate.

With lending power, the bank can therefore prefer to supply less protection and tolerate earnings fluctuations in order to earn higher expected profits. Stable funding gives it a reason to supply protection cheaply, while lending power rewards restricting that supply. This hedging argument applies while $G > 0$: once fixed lending exceeds the full hedge, $G < 0$ and additional protection increases the bank's earnings risk, requiring compensation.

The exogenous r_a affects expected funding costs but, conditional on bank counts, neither marginal incentive in (5).

Proposition 1 (Equilibrium and earnings exposure)**.** *Conditional on the deposit retention equilibrium, there is a unique symmetric lending equilibrium. Let* $D = \theta(1 + 1/N) + \gamma$ *and* $\beta^d = 1 - 1/n$*. Then*

$$L^F = \frac{\theta + \gamma(1 - \beta^d)}{D} = \frac{\theta + \gamma/n}{D}. \tag{6}$$

The corresponding premium is

$$s = \theta\sigma^2 \frac{\theta/N + \gamma\beta^d}{D}, \tag{7}$$

and earnings exposure is $G = \theta[1 - (N + 1)\beta^d]/(ND)$*. The allocation is interior, with* $0 < L^F < 1$ *and* $0 < s < \theta\sigma^2$*. The sensitivity gap is positive, zero, or negative as* β^d *is below, equal to, or above* $1/(N + 1)$*.*

Proof. Combine (2) and (5) to obtain $DL^F = \theta + \gamma(1 - \beta^d) > 0$. The difference between D and this numerator is $\theta/N + \gamma\beta^d > 0$, establishing interiority. Strict own concavity makes the first-order condition sufficient. The symmetric linear equation has one solution, while the symmetric all-zero and all-one portfolios fail the boundary optimality conditions. Substitution gives the premium and the gap. □

Two portfolio targets explain the allocation. Full earnings hedging requires $L_H = 1-\beta^d = 1/n$, while a risk-neutral bank chooses $L_C = N/(N + 1)$. Equilibrium satisfies

$$L^F = \omega L_H + (1 - \omega)L_C, \qquad \omega = \frac{\gamma}{D} \in (0, 1).$$

When $L_H > L_C$, preserving lending rents leads banks to stop short of the full hedge: they retain $G > 0$ and forgo some profit stability. When $L_H < L_C$, earning premium income instead draws them beyond the full hedge, leaving $G < 0$. In both cases, banks accept earnings risk to raise expected profits relative to the perfectly hedged portfolio. Greater risk aversion raises the weight on L_H, but complete hedging occurs only when the two targets coincide.

Greater rate uncertainty σ raises the premium but leaves the fixed share unchanged. Under mean–variance preferences, borrowers' willingness to pay and banks' compensation for bearing risk both scale with variance. Other preferences or endogenous participation (Section 3.6) can change the share response.

3.3 Competition and the premium–beta relationship

Deposit and lending competition change different sides of this tradeoff. More responsive funding reduces the fixed-rate share needed to stabilise earnings, weakening banks' willing-

ness to supply borrower protection. More lenders instead reduce each bank's incentive to restrict supply to preserve the premium.

First vary deposit pass-through and lending competition separately. Although β^d depends on n in equilibrium, treating it as exogenous lets us isolate the funding effect. Holding the other margin fixed gives

$$s_{\beta^d} = \frac{\theta\gamma\sigma^2}{D} > 0, \qquad s_N = -\frac{\theta^2\sigma^2[\theta + \gamma(1-\beta^d)]}{N^2D^2} < 0.$$

Since $d\beta^d/dn = 1/n^2 > 0$, more local deposit competition alone raises the premium. Therefore, *a negative premium–beta relationship requires lending competition to increase enough to offset this funding effect.*

To study joint changes in deposit and lending competition, hold fixed the ratio of lenders to local deposit competitors,

$$M \equiv \frac{N}{n} \geq 1, \qquad N = Mn = \frac{M}{1-\beta^d}.$$

Here M is the number of local deposit markets accessible to borrowers, held fixed in this comparison.

Proposition 2 (Market breadth and the premium slope)**.** *Along symmetric configurations with a fixed ratio $M = N/n$, the fixed premium is decreasing in deposit beta if*

$$M = \frac{N}{n} < \frac{\theta^2}{\gamma(\theta+\gamma)},$$

increasing if the inequality is reversed, and constant at equality. Under the uniform sufficient condition in Appendix C.3, the sign holds over $\beta^d \in [0,1)$ and orders the corresponding integer-count configurations.

Proof. Writing $g = \gamma/\theta$, substitution into (7) gives

$$\frac{s(\beta^d; M)}{\theta\sigma^2} = \frac{1 + (gM-1)\beta^d}{1 + M(1+g) - \beta^d}. \tag{8}$$

Differentiating (8) at fixed M yields

$$\frac{ds}{d\beta^d} = \theta\sigma^2 \frac{M[gM(1+g)-1]}{[1+M(1+g)-\beta^d]^2}.$$

The denominator is positive, giving the result. □

When borrowers rely on a narrow lending market, entry can substantially reduce the rents banks preserve by restricting fixed lending. The resulting expansion of protection can

outweigh the loss of the hedging incentive as funding becomes more rate-sensitive. When borrowers already access many lenders, there is less lending power left to erode, so the funding effect dominates. A negative slope is feasible for $M \geq 1$ only if $\gamma/\theta < (\sqrt{5}-1)/2$. Figure 4 illustrates both regimes; the premium itself remains positive.

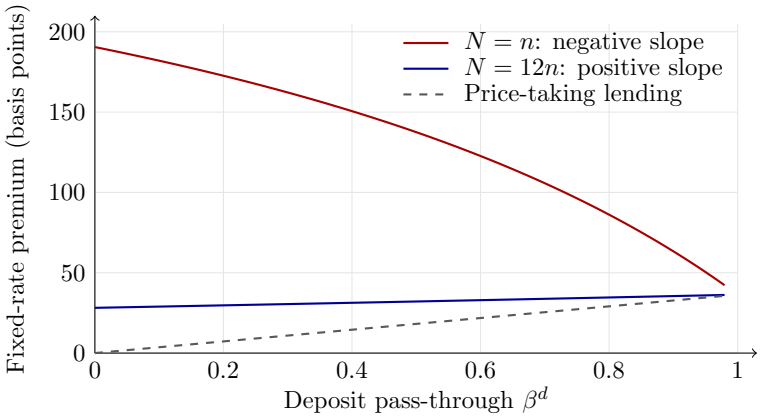


Figure 4: Fixed-rate premiums and deposit pass-through. The illustration uses $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, and $r_0 - r_a = 0.03$, giving a breadth threshold of 9.09 and satisfying the deposit-pricing condition in Appendix C.3. Solid curves hold N/n fixed and vary the continuous competition indices; the dashed curve takes $N \to \infty$ at a fixed beta. The curves are theoretical illustrations, not estimates.

Price-taking lending. Let $N \to \infty$ while holding n, and hence deposit passthrough β_d, fixed. This case has no lending market power. The Cournot term in (5) disappears and (7) implies

$$s_\infty(\beta^d) = \frac{\theta\gamma\sigma^2}{\theta+\gamma}\beta^d.$$

Without the incentive to restrict supply, stable funding translates into cheaper protection: the premium falls to zero as $\beta^d \to 0$. For positive beta, a positive premium compensates banks for taking fixed lending beyond their full hedge. This limit holds deposit competition fixed; taking $n \to \infty$ at fixed M also sends beta to one. Deposit rents can survive price-taking lending because banks choose the composition of predetermined books.

3.4 The premium–beta relationship under CARA and CRRA

The fixed-rate premium can fall as deposit pass-through rises under both constant absolute and constant relative risk aversion. As in Section 3.3, more responsive deposit costs weaken a bank's incentive to hedge with fixed-rate loans. But when borrowers gain access to more lenders, each bank also has less power to sustain a high premium. To establish when lending competition dominates, keep the same deposit contracts and borrower types and let borrowers and banks maximise expected utility.

With a slight abuse of notation, retain $\tilde{\theta} \sim U[0, 2\theta]$ for borrower risk aversion and γ for bank risk aversion. For a borrower set $\varrho = \tilde{\theta}$ and for a bank set $\varrho = \gamma$. Consider two utility functions over terminal wealth w:

$$u_\varrho^{\text{CARA}}(w) = \frac{1 - e^{-\varrho w}}{\varrho}, \qquad u_\varrho^{\text{CRRA}}(w) = \begin{cases} (w^{1-\varrho} - 1)/(1 - \varrho), & \varrho \neq 1, \\ \log w, & \varrho = 1. \end{cases}$$

Under constant absolute risk aversion (CARA), ϱ measures aversion to a given dollar risk. Under constant relative risk aversion (CRRA), it measures aversion to a given proportional risk, and w must be positive. The parameters therefore have different units across specifications. In either case $\varrho = 0$ gives linear utility, taking the limit for CARA.

As before, the most risk-averse borrowers choose fixed rates. At a fixed share L, the marginal borrower has type $2\theta(1 - L)$. Let $c = y - r_0 > \sigma$ be its consumption after paying the expected variable rate. Inverse demand $s(L, \sigma)$ is the premium at which this borrower is indifferent between consuming $c - s$ for certain and consuming $c - \sigma$ or $c + \sigma$ with equal probability.

For both preferences, let the bank start with the same predetermined equity E per unit of assets and evaluate terminal equity $E + \pi_i(z)$. At a symmetric allocation its earnings exposure is $G = 1 - \beta^d - L$, and its terminal equity per unit of assets is $W \pm \sigma G$, where $W = E + (1 - \beta^d)(r_0 - r_a) + Ls$. Under CARA, adding the same equity in both states leaves their relative marginal utilities unchanged. Under CRRA, a given earnings fluctuation matters more when equity is low. We choose E so that terminal equity is positive in both states for every feasible portfolio; $E = 0$ also suffices if earnings alone meet this condition.

Lemma 2 (Fixed-rate pricing under expected utility)**.** *In an interior symmetric Cournot equilibrium, the fixed-rate premium satisfies*

$$s = \underbrace{-\frac{L}{N} s_L}_{\text{Cournot term}} - \underbrace{H}_{\text{marginal hedging benefit}}, \qquad H = \sigma \frac{u'_\gamma(W - \sigma G) - u'_\gamma(W + \sigma G)}{u'_\gamma(W - \sigma G) + u'_\gamma(W + \sigma G)}, \tag{9}$$

where $s_L = \partial s(L, \sigma)/\partial L$ and u_γ is the bank's utility under the specification considered.

Proof. See Appendix C.4. □

Consider a risk-neutral bank taking rivals' quantities as given. An additional fixed loan earns the premium s but lowers the premium on the bank's other fixed loans. Since $\partial L/\partial l_i^F = 1/N$, its first-order condition is $s + (l_i^F/N)s_L = 0$. At the common equilibrium share L_C, this becomes

$$s(L_C, \sigma) + \frac{L_C}{N} s_L(L_C, \sigma) = 0. \tag{10}$$

Now consider a risk-averse bank at that same portfolio. If $L_C < 1 - \beta^d$, its earnings are lower in the low-rate state. An additional fixed loan shifts income into that state. Its marginal expected profit is zero at L_C, but its hedging benefit is positive, so the bank wants to expand fixed lending. If $L_C > 1 - \beta^d$, the same loan adds to earnings risk and the bank wants to supply less. Under CARA the earnings gap determines the value of this hedge. Under CRRA a bank with less equity values the same reduction in risk more.

Proposition 3 (Portfolio choice under expected utility)**.** *Let L_C denote the symmetric Cournot fixed share with risk-neutral banks that solves* (10)*, and $L_H = 1 - \beta^d$ the share that fully hedges bank earnings. Under CARA the symmetric Cournot equilibrium is unique. Its fixed share lies between L_C and L_H, strictly so when they differ. The same result holds under CRRA if premium revenue is strictly concave in the bank's own fixed share, symmetric marginal revenue is strictly decreasing in L, and G/W decreases in L.*

Proof. See Appendix C.4. □

To compare markets, again hold $M = N/n$ fixed. As deposit beta rises, the bank needs fewer fixed loans to hedge deposits. At the same time, borrowers gain access to more lenders, reducing the premium revenue each bank gives up when it expands fixed lending.

Proposition 4 (Market breadth and the premium slope under expected utility)**.** *Along $N = Mn$, suppose the marginal return $s + (L/N)s_L + H$ is strictly downward sloping in L at the symmetric equilibrium. The fixed-rate premium decreases with deposit beta if and only if*

$$\frac{L(-s_L)}{M} > -\left.\frac{\partial H}{\partial \beta^d}\right|_L.$$

Under the conditions of Proposition 3, if symmetric marginal revenue has strictly negative slope at L_C, this inequality holds for sufficiently small positive γ under either utility function.

Proof. See Appendix C.4. □

The left side measures how much entry weakens the incentive to restrict fixed lending. The right side measures how much more responsive deposits weaken the incentive to hedge. When banks place little weight on earnings risk, the pricing incentive dominates even though banks still value a hedge.

In the CRRA illustration, doubling n from two to four raises deposit beta from 0.50 to 0.75. Only market breadth differs across economies. Set $\theta = 2$ (borrower types range from zero to four), $\gamma = 0.80$, $c = 0.08$, $E = 0.10$, and $r_0 - r_a = 0.03$. A larger shock, $\sigma = 0.02$, makes the premium changes easier to see.

With two deposit markets ($M = 2$), borrowers gain access to eight lenders instead of four ($N = Mn$). The premium falls from 25.66 to 24.50 basis points and the fixed share rises

from 74.75% to 75.89%. With twelve deposit markets ($M = 12$), borrowers already face 24 lenders before the increase and 48 afterwards. The premium rises from 13.73 to 18.45 basis points and the fixed share falls from 86.49% to 81.86%.

Where borrowers initially have few lenders, competition makes protection cheaper despite weaker hedging incentives. In a broad market, lenders have less pricing power to lose and the funding effect can raise the premium.

3.5 Welfare and the allocation of interest-rate risk

Deposit competition determines how closely funding costs follow market rates. A bank with stable deposit costs can insure borrowers against a rate increase at little risk to its earnings. Lending competition determines how much of that protection reaches borrowers: selling another fixed loan requires a bank with pricing power to lower the premium on its other fixed loans.

Compare efficient and Cournot lending for the same deposit contracts, bank counts, loan menu, and total credit. Premium payments transfer income from borrowers to banks. The loss arises when a borrower retains rate risk even though fixing the loan would raise joint surplus.

Let L be the share of loans allocated to fixed-rate borrowers, starting with the most risk averse. Borrowers with $\tilde{\theta} < 2\theta(1 - L)$ retain variable-rate loans, and each bank retains an earnings exposure of $1 - \beta^d - L$. Summing borrower and depositor certainty equivalents and weighting banks' per-asset certainty equivalents by their asset shares gives, up to a constant,

$$\mathcal{W}(L) = -\frac{\sigma^2}{2}\left[\theta(1 - L)^2 + \gamma(1 - \beta^d - L)^2\right]. \tag{11}$$

The first term measures the cost of rate risk borne by borrowers and the second the cost borne by banks. Risk-neutral depositors require no compensation for risk. The efficient share,

$$L^\star = \frac{\theta + \gamma(1 - \beta^d)}{\theta + \gamma},$$

gives more weight to insuring borrowers when θ is high and to hedging bank earnings when γ is high. At the Cournot allocation, $\mathcal{W}'(L^F) = \theta\sigma^2 L^F/N > 0$: the gain from one more fixed loan equals the premium revenue the bank would forgo on its other fixed loans. By (11), the loss from this shortfall is

$$\mathcal{W}(L^\star) - \mathcal{W}(L^F) = \frac{\sigma^2}{2}(\theta + \gamma)(L^\star - L^F)^2.$$

Under mean–variance preferences, greater uncertainty raises both parties' risk costs in proportion to σ^2. The efficient and equilibrium shares therefore stay fixed, and doubling σ

quadruples the welfare loss for any γ/θ.

Uncertainty and risk aversion. With CARA preferences, a more risk-averse borrower will pay more to fix the rate. What makes it profitable for a bank to withhold a loan, however, is the difference between what it can charge its current borrowers and what it must charge to attract the next one. For small shocks, marginal willingness to pay is approximately $\theta\sigma^2(1-L)$. A higher θ then steepens inverse demand: expanding fixed lending requires a larger premium cut on the bank's existing fixed volume. This is one way borrower risk aversion can make lending power more costly.

Proposition 5 (Uncertainty and the welfare cost of lending power)**.** *Under CARA preferences with $\theta, \gamma > 0$, fix deposit pass-through $0 < \beta^d < 1$ and a finite number of lenders. Let $\Delta(\sigma)$ be the gain in aggregate monetary certainty equivalents from replacing symmetric Cournot lending with efficient lending. The gain is positive for every $\sigma > 0$ and attains its maximum at an interior level of uncertainty. It tends to zero as $\sigma \to 0$ and as $\sigma \to \infty$.*

Proof. See Appendix C.5. □

The loss therefore has an interior peak. At low uncertainty, an increase in σ makes each loan left unhedged more costly. At high uncertainty, even the marginal fixed-rate borrower will pay close to σ for protection. Borrowers' valuations converge relative to the size of the shock, while banks place increasing weight on losses in their adverse state. Both allocations approach the bank's full hedge, $1-\beta^d$, and the additional loss due to lending power vanishes. This large-shock comparison holds deposit contracts fixed.

For $n = N = 2$ and $\beta^d = 1/2$, raise σ from 0.005 to 0.015. At $\theta = 1600$, the loss rises from 2.74 to 4.14 basis points per unit lent for $\gamma = 80$ but falls from 1.24 to 0.46 for $\gamma = 320$. Borrower preferences can also reverse the comparison. At $\gamma = 160$, the loss rises from 2.08 to 2.12 basis points for $\theta = 800$ but falls from 2.11 to 1.91 for $\theta = 1600$, even though these borrowers value protection more.

3.6 Market size and endogenous entry

Section 3.3 shows when the premium falls as bank counts rise. What makes depositors and borrowers face more competitors in the first place? A larger customer base can cover the fixed cost of another bank. Because each entrant takes deposits locally and lends across communities, one participation decision expands choice on both sides of the market.

Let $A > 0$ denote both the mass of unit loans and total deposits. Each of the M communities supplies A/M deposits. Search remains uncongested, so market size affects pass-through only through entry. For this extension assume $r_0 - r_a > \gamma\sigma^2$, which rules

out voluntary extra pass-through after any portfolio choice and supports the franchise-value inequalities below. Earlier results require only the weaker condition in Appendix C.3.

To operate, owners pay $F > 0$ in terminal goods from outside resources. This cost is independent of market size and beta. Staying out pays zero. Before search, active banks share home-community deposits equally, extending the benchmark allocation without a separate franchise acquisition price. We study configurations with at least one bank per community.

Under symmetry, $N = Mn$ banks each hold $A/(Mn)$ in assets. Fixed volume is AL^F, so inverse demand (2) and all per-asset outcomes carry over. Risk-adjusted value per unit of assets before entry costs is

$$W(n; M) = \frac{r_0 - r_a}{n} + L^F(n; M)s(n; M) - \frac{\gamma}{2}\sigma^2 \left[\frac{1}{n} - L^F(n; M)\right]^2, \tag{12}$$

and $v(n; M) = W(n; M)/(Mn)$ gives bank value per unit of market size. Entry yields $Av(n; M) - F$. At given counts a higher outside return leaves portfolios unchanged but erodes deposit rents, $\partial W/\partial r_a = -1/n$. We hold r_a fixed.

We use a continuous bank count to trace how participation responds to market size. Appendix C.7 gives the exact entry conditions for integer counts in one or several communities.

Proposition 6 (Market size and participation)**.** *Under the maintained parameter restrictions, $W(n; M) > 0$ and $W_n(n; M) < 0$ for every real $n \geq 1$ and $M \geq 1$. Consequently $v(n; M)$ is strictly decreasing and tends to zero.*

For any fixed $M \geq 1$, the continuous symmetric participation equation

$$\frac{A}{Mn}W(n; M) = F \tag{13}$$

has a unique solution $n > 1$ whenever $AW(1; M)/M > F$. Along this approximation,

$$\frac{dn}{dA} = \frac{n/A}{1 - nW_n/W} > 0, \qquad \frac{d\beta^d}{dA} = \frac{1}{n^2}\frac{dn}{dA} > 0, \qquad \frac{dN}{dA} = M\frac{dn}{dA} > 0. \tag{14}$$

The premium–beta slope is again given by Proposition 2.

Proof. See Appendix C.6. □

Entry divides the larger market among more banks. More depositors receive competing offers and more lenders compete for fixed-rate borrowers. Both margins erode rents, limiting further entry.

At fixed M, market size thus selects a point on the premium–beta curve in Section 3.3. Larger markets have higher beta and cheaper protection when competition for borrowers dominates. When more responsive funding matters more for hedging, beta and the premium rise together. Proposition 2 gives the threshold.

4 Loan pricing, investment, and the macroeconomy

We now let entrepreneurs choose project size as well as loan type. More expensive fixed-rate protection induces both a switch to floating finance and lower investment. We first establish this link, then embed bank pricing and entry in an equilibrium with household saving and endogenous wages, conditional on the benchmark and outside rates.

Environment. There are M markets. Market m has a mass φ_m of entrepreneurs, with $\sum_{m=1}^{M} \varphi_m = 1$. As in Section 3.6, we use a continuous count n_m for active banks and write $N = \sum_m n_m$. Here each bank takes deposits and lends in its own market. Borrowers access a subset of local lenders, as specified in Section 4.2. Depositors search as in (3), with local contact rate ζ_m relative to the decision deadline. Under the same deposit-pricing rule,

$$r_m^d(z) = (1 - \beta_m^d) r_a + \beta_m^d r_1(z), \qquad \beta_m^d = p(n_m; \zeta_m). \tag{15}$$

The real interbank rate follows (1). We vary r_0 holding σ, r_a, search frictions and participation cost schedules fixed. Appendix D.2 describes implementation of the benchmark rate.

Labour moves freely across markets, so all entrepreneurs face a common wage. Retailers contract their date-0 supply prices before the change in r_0 and price flexibly at date 1. Banks observe r_0 and decide whether to serve a market before deposit search and lending. Loan terms, investment and employment at both dates are chosen at date 0, before the rate shock is realised. The terminal wage w_1 clears the market for committed date-1 labour at date 0: it adjusts to changes in r_0, but not to the subsequent rate realisation. Contracts settle at date 1.

4.1 Borrower investment and loan choice

Entrepreneurs sell intermediate output at $\bar{p} = (\varepsilon - 1)/\varepsilon$, where $\varepsilon > 1$ is the retail substitution elasticity. Each borrows the full cost of investment k, hires h hours of labour and produces $A_1 k^\alpha h^{1-\alpha}$ intermediate goods, where $A_1 > 0$ is productivity and $\alpha \in (0, 1)$ is the capital share. Capital depreciates fully and expansion costs $\kappa k^2/2$ terminal goods, with $\kappa > 0$. In market m, borrower risk aversion satisfies $\log \tilde{\theta}_m \sim N(\mu_m, \tau^2)$, where μ_m is the market-specific mean of log risk aversion and $\tau > 0$ is its common standard deviation. Subscripts F and V denote fixed- and variable-rate finance. Given loan rates and the wage, an entrepreneur chooses contract $j \in \{F, V\}$, investment and committed hours before the rate shock to solve

$$\max_{j \in \{F,V\},\, k,h \geq 0} -\mathbb{E}\Big[\exp\Big(-\tilde{\theta}_m \Pi_j(z)\Big)\Big], \tag{16}$$

Here $z \in \{-1, 1\}$ is the equally likely rate-shock state, and $\mathbb{E}$ averages over these two states. An argument z indicates a state-contingent payoff or rate; the choices j, k, h are made before z is known. The payoff $\Pi_j(z)$ includes any profit-equivalent contract benefit, with dependence on k, h and market m suppressed. Cash profit is output revenue less wages, loan repayment and expansion costs. The loan rates are $r_F = r_0 + s_m$ and $r_V(z) = r_0 + \sigma z$, where s_m is the local fixed-rate premium and $\sigma > 0$ is the size of the rate shock. Fixed-rate profit is independent of z, so we write it as Π_F. Optimal hours depend on the wage and project size:

$$h = \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{1/\alpha} k.$$

Substituting labour demand gives

$$\Pi_F = (R - s_m + v_F)k - \frac{\kappa}{2}k^2, \qquad \Pi_V(z) = (R - \sigma z)k - \frac{\kappa}{2}k^2.$$

Here $v_F k$ is a nonpecuniary convenience benefit measured in profit-equivalent units, for example from simpler payment administration. Its additive, proportional form assumes constant willingness to pay per unit financed, separate from the insurance value captured by risk aversion. The common expected surplus is

$$R = \alpha\bar{p}A_1 \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{(1-\alpha)/\alpha} - 1 - r_0. \tag{17}$$

Thus R is the expected surplus per unit invested after wages, principal and benchmark interest, before the fixed premium and expansion costs.

Write k_F and $k_V(\tilde{\theta}_m)$ for optimal investment under fixed- and variable-rate finance, suppressing the market index. Both are chosen before z is realised. On the interior branch, $k_F = (R - s_m + v_F)/\kappa$. With variable finance, each extra unit borrowed increases exposure to the rate shock, and optimal investment solves

$$\kappa k_V(\tilde{\theta}_m) + \sigma \tanh[\tilde{\theta}_m \sigma k_V(\tilde{\theta}_m)] = R.$$

Appendix D.1 derives the investment conditions. Borrowers compare the two contracts at their respective optimal scales. The cutoff θ_m^* equates their certainty equivalents, and types above it fix. A higher premium reduces the size of fixed-rate projects and leads some borrowers to switch contracts.

Let F_m^θ denote the cumulative distribution of borrower risk aversion. Then $f_m^F = 1 - F_m^\theta(\theta_m^*)$ is the fraction choosing fixed finance, with $\theta_m^* = \theta^*(s_m; R)$. Superscripts F, V on loan volumes likewise identify the contract type. Fixed lending equals borrower mass times

fixed take-up times loan size:

$$Q_m^F = \varphi_m f_m^F \frac{R - s_m + v_F}{\kappa}.$$

Variable loan sizes vary across borrower types t, so $Q_m^V = \varphi_m \int_0^{\theta_m^*} k_V(t)\, \mathrm{d}F_m^\theta(t)$. Together these loans finance local investment $I_m = Q_m^F + Q_m^V$.

Remark: At a given wage and benchmark rate, the indifferent borrower invests more under fixed finance: $k_F > k_V(\theta_m^*)$, as proved in Appendix D.1. A higher premium therefore reduces investment both among borrowers who remain fixed and among those who switch to floating. Along borrower demand,

$$-1 < \frac{\partial Q_m^V}{\partial Q_m^F} < 0, \qquad \frac{\partial I_m}{\partial s_m} < 0. \tag{18}$$

Floating credit replaces less than one-for-one the reduction in fixed credit. Appendix D.1 derives these inequalities; equilibrium wage adjustment can moderate the investment response.

4.2 Bank pricing and participation

Bank i operates in market m, taking deposits and lending locally. Household deposits $\mathcal{D}$ are allocated in fixed shares δ_m, so local deposits are $\mathcal{D}_m = \delta_m \mathcal{D}$ and each bank receives $d_m = \mathcal{D}_m / n_m$. Its fixed- and variable-rate loan volumes are q_{im}^F and q_{im}^V. Net interbank lending x_{im} is positive when the bank lends and negative when it borrows, with

$$q_{im}^F + q_{im}^V + x_{im} = d_m.$$

Variable loans and interbank positions earn $r_1(z)$; only variable loans generate net ancillary-service income b_V per unit. Banks have no outside fixed-rate hedge. Equity is $E_{im} = e d_m$, where $e > 0$ is a common equity-to-deposit scale used to normalise risk bearing, not an additional funding flow.

Borrowers can approach only banks in their lending segment, while all n_m banks compete for deposits across market m. This distinction is motivated by the role of branch networks and local competition in credit access (Aguirregabiria et al., 2025).[6] With J identical segments, each contains φ_m / J borrowers with the same type distribution and n_m / J banks. Segment inverse demand is $s_m(J Q_{m\ell}^F; R)$, where $Q_{m\ell}^F$ is segment fixed lending. At symmetry, its slope is $J s_{m,Q}$, where $s_{m,Q} = \partial s_m(Q_m^F; R)/\partial Q_m^F$. We extend these symmetric conditions continuously to $J \geq 1$, interpreted as effective segmentation: larger values mean

[6] Singh and Vives (1984) study price and quantity competition with differentiated products, while Weyl and Fabinger (2013) develop a general framework for imperfect competition and pass-through. Here the pricing multiplier arises from restricted access to otherwise identical lenders under Cournot competition within segments; borrower preferences over available contracts are unchanged.

fewer accessible lenders, n_m/J (Appendix D.1).

Taking funding, wages and rivals' fixed quantities within its segment as given, bank i chooses fixed lending before z is realised. With common CARA coefficient $\gamma_E > 0$ over return on equity, its problem is

$$\max_{q^F_{im} \geq 0} -\mathbb{E}\left[\exp\left(-\gamma_E \frac{\Pi_{im}(z)}{E_{im}}\right)\right], \tag{19}$$

where s_m denotes its segment premium (common across segments at symmetry) and operating profit before fixed overhead is

$$\Pi_{im}(z) = d_m(1-\beta^d_m)[r_1(z) - r_a] + q^F_{im}[r_0 + s_m - r_1(z)] + b_V q^V_{im}.$$

Changes in segment variable lending accrue to the bank changing fixed lending; interbank positions clear its balance sheet, with $q^V_{im} \geq 0$. Thus $\partial q^V_{im}/\partial q^F_{im} = Q^V_{m,Q}$, the demand derivative in (18).

At a symmetric allocation, differentiating the bank's certainty equivalent gives the interior condition

$$0 = s_m + J q^F_{im} s_{m,Q} + b_V Q^V_{m,Q} + \sigma \tanh\left[\frac{\gamma_E \sigma}{E_{im}}\left(d_m(1-\beta^d_m) - q^F_{im}\right)\right].$$

The terms are marginal premium revenue, the change in variable-loan service income, and the hedging benefit (Appendix D.1). Under symmetry, $q^F_{im} = Q^F_m/n_m$ and $d_m = \mathcal{D}_m/n_m$, giving

$$s_m = -\frac{JQ^F_m}{n_m} s_{m,Q} - b_V Q^V_{m,Q} - \sigma \tanh\left[\frac{\gamma_E}{e}\sigma\left(1 - \beta^d_m - \frac{Q^F_m}{\mathcal{D}_m}\right)\right].$$

Banks trade premium and service income against the value of stabilising earnings: fixed credit provides a hedge when asset income is more rate-sensitive than deposit costs.

Entry is market-specific. In each market, the symmetric bank's profit certainty equivalent equals the local operating hurdle $F_m > 0$:

$$-\frac{E_{im}}{\gamma_E} \log \mathbb{E}\left[\exp\left(-\gamma_E \frac{\Pi_{im}(z)}{E_{im}}\right)\right] = F_m. \tag{20}$$

The hurdle includes fixed overhead in terminal goods and can differ across markets. Free entry makes net entry value zero in each active market; we impose no additional condition equating gross bank values across markets.

A higher r_0 widens deposit margins and encourages entry. More lenders reduce the premium income sacrificed when expanding fixed credit, but higher deposit pass-through weakens the incentive to hedge. These opposing forces determine the price of protection.

4.3 Households

A representative household chooses current consumption C_0, deposits $\mathcal{D}$, hours L_t at dates $t = 0, 1$, and terminal consumption $C_1(z)$. Deposits and hours are chosen at date 0; terminal consumption depends on the rate-shock state z. Taking wages, deposit returns and retail dividends as given, it solves

$$\max_{C_0,\mathcal{D},L_0,L_1,\{C_1(z)\}} \left\{ \log C_0 - \frac{\chi_0 L_0^{1+\nu_0}}{1+\nu_0} + \delta_H \left(\mathbb{E}[C_1(z)] - \frac{\chi_1 L_1^{1+\nu_1}}{1+\nu_1} \right) \right\}, \tag{21}$$

subject to the budgets below and nonnegative choices, with $C_0 > 0$. Here $\delta_H \in (0,1)$ is the discount factor, $\chi_t > 0$ scales labour disutility, and $\nu_t > 0$ determines its curvature. Expectations average over the two rate states. Linear terminal utility makes depositors rank returns by expected repayment.

Let w_t denote the common real wage and Ω_t^R retail dividends at date t. Fixed deposit relationship shares δ_m sum to one and need not equal borrower shares φ_m. The household's gross deposit return is $R^d(z) = \sum_m \delta_m[1 + r_m^d(z)]$. Its budgets are

$$C_0 + \mathcal{D} = w_0 L_0 + \Omega_0^R. \tag{22}$$

$$C_1(z) = w_1 L_1 + R^d(z)\mathcal{D} + \Omega_1^R, \qquad z \in \{-1, 1\}. \tag{23}$$

The household owns retailers; entrepreneurs and bank owners consume their own profits. At an interior optimum, an extra unit saved costs $1/C_0$ in current marginal utility and yields discounted expected repayment $\delta_H \mathbb{E}[R^d(z)]$, giving

$$\frac{1}{C_0} = \delta_H \, \mathbb{E}[R^d(z)]. \tag{24}$$

Labour supply equates marginal disutility to wage income valued at marginal utility of consumption:

$$\chi_0 L_0^{\nu_0} = w_0/C_0, \qquad \chi_1 L_1^{\nu_1} = w_1.$$

Terminal marginal utility is one, so the common discount factor cancels from the date-1 labour condition. A higher expected deposit return lowers C_0 at an interior solution. The terminal wage clears the market for committed hours at date 0.

4.4 Aggregation and equilibrium

Credit and funding. Local investment is $I_m = Q_m^F + Q_m^V$, the sum of fixed- and variable-rate lending. With no outside asset, aggregate investment I equals household deposits:

$$I = \sum_{m=1}^{M} I_m = \mathcal{D}. \tag{25}$$

Net interbank lending from market m is $X_m = n_m x_{im} = \mathcal{D}_m - I_m$. The closed interbank market requires $\sum_m X_m = 0$: local funding surpluses finance other markets' deficits. The aggregate fixed share of loan volume is $L^{F,\text{agg}} = \sum_m Q_m^F / I$, while borrower take-up is $f^{F,\text{agg}} = \sum_m \varphi_m f_m^F$.

Production and labour. Competitive date-0 producers have productivity A_0, produce aggregate output $Y_0 = A_0 L_0$, and earn zero profits at intermediate-good price $p_0 = w_0/A_0$. At date 1, all projects face the same wage and therefore use the same hours per unit invested. Aggregate labour demand and output are

$$L_1 = \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{1/\alpha} I, \qquad Y_1 = A_1 I^{\alpha} L_1^{1-\alpha}. \tag{26}$$

Together with household labour supply, these conditions determine the common wages and hours committed at date 0.

Monopolistically competitive retailers convert intermediate goods one for one into final consumption and investment goods. With the final-good price normalised to one, flexible terminal pricing gives $p_1 = \bar{p} = (\varepsilon - 1)/\varepsilon$. Preset current retail prices allow p_0 to adjust after a rate change. Retail dividends at date $t \in \{0, 1\}$ are

$$\Omega_t^R = (1 - p_t) Y_t. \tag{27}$$

Appendix D.2 derives pricing and the baseline condition $p_0 = \bar{p}$.

Goods markets. Current output finances household consumption and investment:

$$Y_0 = C_0 + I. \tag{28}$$

Let $C_1^E(z)$ and $C_1^B(z)$ denote terminal consumption of entrepreneurs and bank owners; $C_1(z)$ is household consumption. Write $J_2 = \sum_m \varphi_m \int_0^\infty k_m(\theta)^2 \, \mathrm{d}F_m^\theta(\theta)$ for the aggregate second moment of project size, where $k_m(\theta)$ is investment chosen by type θ under its preferred

contract. In each state,

$$Y_1 + b_V \sum_m Q_m^V = C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2 + \sum_m n_m F_m, \quad z \in \{-1, 1\}. \tag{29}$$

Net ancillary-service output $b_V \sum_m Q_m^V$ adds to project output; expansion costs $\kappa J_2/2$ and bank overhead $\sum_m n_m F_m$ use resources. Loan and deposit repayments transfer resources across owners and households. The nonpecuniary benefit $v_F k$ is excluded from goods accounts. Appendix D.3 verifies this identity state by state.

Definition 1 (General equilibrium conditional on the benchmark and outside rates)**.** *Given the rate process, outside return, borrower populations, deposit shares, segmentation J, search frictions, entry hurdles and inherited retail contracts, an interior equilibrium consists of local bank counts, prices, wages, consumption, employment, investment and financial portfolios such that:*

1. *The household maximises* (21) *subject to* (22) *and* (23)*, taking prices, dividends and deposit shares as given.*

2. *Entrepreneurs choose contract, scale and committed employment. Producers, retailers and the final-good aggregator optimise at inherited current retail prices and flexible terminal prices.*

3. *In each market $n_m > 1$, with $n_m/J \geq 1$, satisfies* (20) *at the anticipated lending equilibrium. Given n_m, deposit rates follow* (15) *and lending satisfies the restricted-access pricing condition, with $q_{im}^F = Q_m^F/n_m$ and $q_{im}^V = Q_m^V/n_m$.*

4. *Bank balance sheets and loan markets clear, interbank positions sum to zero, and* (25)*,* (28)*,* (26) *and* (29) *clear funding, goods and labour markets. Consumption and repayments are feasible in both states.*

Under symmetry each lender originates a share $1/n_m$ of market-wide credit, so the measured lending Herfindahl index is $1/n_m$. Concentration among the lenders accessible within a segment is instead J/n_m.

5 Quantitative results

We calibrate the joint lending, household saving and investment equilibrium of Section 4. Households choose aggregate deposits, entrepreneurs choose both loan contracts and project sizes, and banks enter local markets. The policy-rate experiment and the risk-bearing-capacity application use the same parameters and preferences.

5.1 Specification and calibration

We calibrate three markets with low, middle and high deposit betas. Each holds one third of deposits, while entrepreneur shares are estimated subject to $\varphi_1 \geq \varphi_2 \geq \varphi_3$. Table 2 reports the parameters and their calibration targets.

Baseline lending concentration determines the number of banks, and deposit betas then determine search frictions. For market m, with baseline HHI H_m^b,

$$n_m^b = 1/H_m^b, \qquad \zeta_m = \frac{\beta_m^{d,b}}{(1-\beta_m^{d,b})(n_m^b - 1)}.$$

The HHI measures concentration across the whole market; borrowers have access to n_m/J lenders. We jointly choose bank risk aversion γ_E, borrower dispersion τ, segmentation J, the benefits b_V, v_F, three borrower distribution means μ_m, and two independent entrepreneur shares to match baseline premiums and fixed-loan shares in each market and four responses to a 50 bp rate increase. These responses are the aggregate changes in the fixed share, premium and HHI, and the high-beta market's fixed-share change. Borrower preferences mainly affect demand for protection, bank parameters affect its supply, and entrepreneur shares determine local credit demand and aggregation weights. Because these channels interact, the parameters are fitted jointly rather than assigned individual targets. The estimate $J = 3.13$ uses the continuous approximation in Appendix D.1. Entry hurdles F_m follow from baseline free entry and remain fixed after the shock.

The remaining assumptions and normalisations are also listed in Table 2. The real benchmark rate is -0.75%, approximately the nominal return of 1.31% less expected inflation of 2.06%. The 59 bp funding–outside-return gap implies $r_a = -1.34\%$. We normalise baseline investment, deposits and hours at both dates to one. This determines baseline surplus R^b jointly with the financial fit; productivity and labour-disutility scales then follow from equilibrium (Appendix D.2).

The response targets refer to the equilibrium after entry has fully adjusted, holding r_a fixed. Premium and HHI changes use baseline loan-volume weights $\omega_m^b = I_m^b/I^b$, whereas aggregate fixed shares use each equilibrium's loan volumes. Investment, deposits, loan quantities, consumption and employment are untargeted predictions, reported in Table 3. Loan quantities reflect both the fitted changes in fixed shares and endogenous investment; real outcomes also depend on the assumed production and labour-supply curvatures. Table 4 shows the market-level responses, identifying the one targeted change. The risk-bearing-capacity exercise uses the same calibration without additional targets.

Table 2: Parameters and calibration targets

Parameter	Value	Target or calibration basis
Jointly fitted financial parameters		
Log dispersion τ	2.94	*Joint targets for these parameters:*
Bank risk aversion γ_E	11.53	Baseline premiums (bp):
Lending segmentation J	3.13	$(23.90, 14.20, 4.30)$;
Bank floating benefit b_V (bp)	12.83	baseline fixed shares (%):
Borrower benefit v_F (bp)	1.41	$(18.60, 18.80, 21.50)$.
Borrower medians e^{μ_m}	(16.05, 8.17, 2.28)	After a 50 bp rate increase:
Shares (φ_1, φ_2) (%)	(53.16, 26.08)	aggregate fixed share +1.30 pp; premium −3.00 bp; HHI −0.067; high-beta fixed share −5.00 pp.
Search and entry parameters		
Deposit search ζ_m	(0.09, 0.15, 0.14)	Baseline betas: $(0.22, 0.35, 0.48)$, given n_m^b
Entry hurdles $10^6 F_m$	(553.04, 330.11, 153.59)	Free entry at baseline HHIs $(0.25, 0.22, 0.13)$
Maintained assumptions		
Equity/deposit ratio e	0.10	Risk-bearing normalisation
Rate uncertainty σ (bp)	50.00	Maintained rate uncertainty
Real financing rate r_0 (%)	-0.75	Nominal benchmark less inflation
Real outside return r_a (%)	-1.34	Implied by the 59 bp rate gap
Rate gap $r_0 - r_a$ (bp)	59.00	Imposed funding–saving gap
Deposit shares δ_m	1/3	Equal deposit relationship shares
Capital exponent α	1/3	Maintained production curvature
Expansion curvature κ	0.10	Maintained investment curvature
Retail elasticity ε	6.00	Maintained retail markup
Discount factor δ_H	0.96	Maintained household preference
Labour exponents (ν_0, ν_1)	(2.00, 2.00)	Maintained labour-supply curvature
Baseline normalisations		
Project productivity A_1	3.94	Baseline $I^b = L_1^b = 1$
Current productivity A_0	2.05	Baseline $L_0^b = 1$ and goods clearing
Labour scale χ_0	1.62	Baseline $p_0^b = (\varepsilon - 1)/\varepsilon$
Labour scale χ_1	2.19	Baseline $L_1^b = 1$ and labour clearing
Baseline surplus R^b	0.10	Baseline $I^b = 1$ with joint fit

Notes: Market-specific values are ordered by low, middle and high deposit beta. The first group contains ten free parameters fitted jointly to ten moments: five common parameters, three borrower medians and two entrepreneur shares. The third share is implied by $\varphi_3 = 1 - \varphi_1 - \varphi_2 = 20.76\%$, with $\varphi_1 \geq \varphi_2 \geq \varphi_3$. All targets are matched at the displayed precision. Premium and HHI responses use baseline loan-volume weights. Borrower distributions are reported by their medians and common log dispersion. Entry hurdles are in profit units. Values are rounded only for presentation.

5.2 Responses to a higher financing rate

Figure 5 traces the equilibrium response to benchmark-rate increases from 0 to 100 bp, holding r_a and all other parameters fixed and allowing entry, saving and investment to

Table 3: Untargeted aggregate responses to a 50 bp rate increase

Outcome	Change (%)
Investment and deposits	−0.67
Fixed lending	+6.03
Floating lending	−2.27
Current consumption	−0.29
Terminal employment	−0.10

Notes: Responses are measured after full entry adjustment. None of these outcomes is targeted. Loan quantities combine fitted composition changes and endogenous investment; real outcomes depend on the maintained production and labour-supply assumptions. Calibration targets are reported in Table 2.

adjust. Diamonds mark the four calibration targets at +50 bp: an aggregate premium decline of 3.00 bp, an aggregate fixed-share increase of 1.30 pp, an aggregate HHI decline of 0.067, and a high-beta fixed-share decline of 5.00 pp. Responses elsewhere on the curves are model implications, with no additional fitting. These are comparative statics after full adjustment, not transition paths.

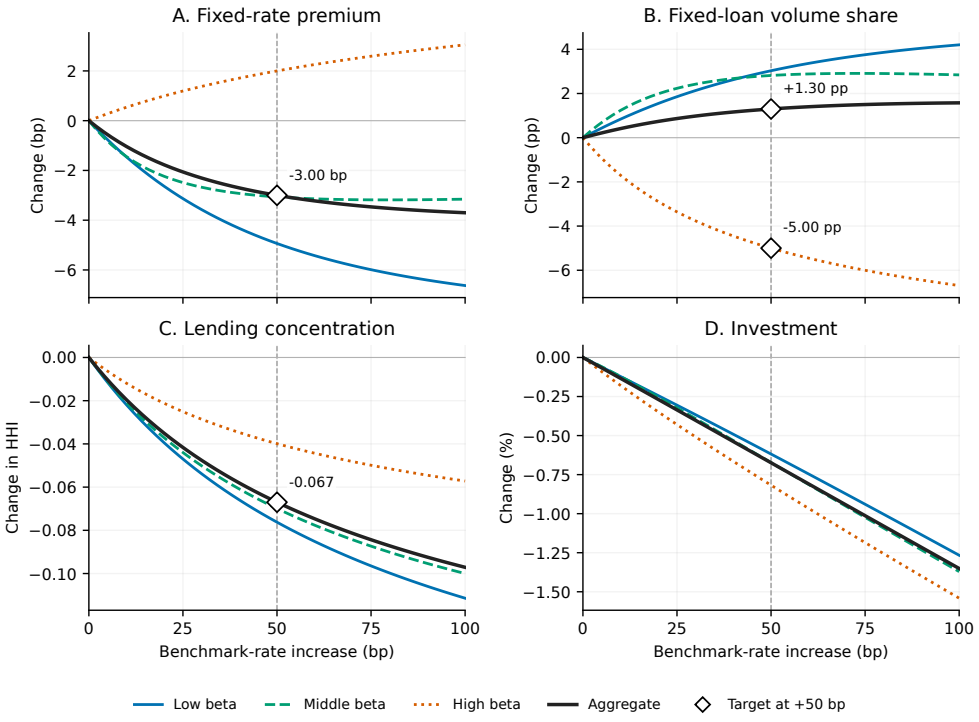


Figure 5: Responses to benchmark-rate increases

Notes: Changes are relative to the baseline. Each point is an equilibrium with fully adjusted entry, saving and investment, holding r_a and all other parameters fixed. Aggregate premiums and HHIs use baseline loan-volume weights; aggregate fixed shares use current loan volumes. Investment changes are percentages of baseline investment. Diamonds mark the targeted aggregate premium, aggregate fixed share, aggregate HHI and high-beta fixed share at a 50 bp increase. Investment is untargeted. The curves are comparative statics, not time paths; all plotted equilibria pass local entry-stability checks.

Responses across rate increases. Panels A–C show how entry and hedging shape loan pricing and composition. At unchanged deposit betas, a rate increase Δr_0 widens deposit margins by $(1-\beta_m^d)\Delta r_0$, encouraging entry and lowering concentration. More lenders reduce banks' price impact in lending, pushing premiums down. Competition for deposits also raises pass-through, reducing the earnings exposure that fixed loans hedge and weakening banks' incentive to supply protection. The equilibrium reflects both forces: premiums fall and fixed shares rise in the low- and middle-beta markets, whereas the high-beta market moves in the opposite direction. The aggregate premium decline and fixed-share increase flatten as the rate increase grows; investment continues to fall in all three markets (Panel D).

Heterogeneity at a 50 bp increase. Table 4 details the cross-market differences at the calibrated shock. Low- and middle-beta premiums fall by 4.94 and 3.07 bp, with fixed shares rising by 3.03 and 2.82 pp. The high-beta premium instead rises by 2.00 bp and its fixed share falls by the targeted 5.00 pp. Individual premium changes and the first two share changes are not separately targeted, although the aggregate targets constrain them. Entry lowers concentration and raises deposit betas in every market, but these changes need not translate into cheaper fixed-rate protection when the hedging incentive weakens.

The shift in loan composition is substantial: fixed lending rises by 15.57% and 14.21% in the first two markets but falls by 23.88% in the third. Total investment falls much less, by 0.62%, 0.67% and 0.82%, respectively. These untargeted quantity responses distinguish substitution between loan contracts from the contraction in overall credit; they are not independent of the fitted share responses.

Table 4: Heterogeneous responses to a 50 bp rate increase

	Low beta	Middle beta	High beta
Post-shock fixed share (%)	21.63	21.62	16.50
Fixed-share change (pp)	+3.03	+2.82	-5.00
Post-shock premium (bp)	18.96	11.13	6.30
Premium change (bp)	-4.94	-3.07	+2.00
Post-shock HHI	0.17	0.15	0.09
HHI change	-0.08	-0.07	-0.04
Post-shock bank count	5.76	6.66	11.10
Post-shock deposit beta	0.31	0.46	0.58
Fixed-loan quantity change (%)	+15.57	+14.21	-23.88
Total investment change (%)	-0.62	-0.67	-0.82

Notes: The high-beta fixed-share decline of 5.00 pp is targeted (Table 2); the other market responses are not separately targeted, although their aggregates are constrained by the calibration targets. Entry adjusts fully. Shares are fractions of loan volume, not borrower counts. Entrepreneur populations remain fixed, while local investment and household deposits adjust.

Aggregate real effects. At +50 bp, the lower premium only partly offsets the benchmark increase: the fixed borrowing rate rises by 47.00 bp using baseline loan-volume weights. Higher financing costs reduce desired project sizes, while wages and saving adjust to clear markets. Table 3 reports a 0.67% decline in investment and deposits, a 0.29% decline in current consumption and a 0.10% decline in terminal employment. Fixed lending nevertheless rises by 6.03% as borrowing shifts away from floating contracts. These real and quantity responses are untargeted and depend on the maintained production and labour-supply assumptions.

5.3 Bank risk-bearing capacity and investment

We use the same baseline calibration to ask how greater bank risk-bearing capacity affects lending. Capacity c rises from 1 to 1.8, replacing γ_E by γ_E/c in banks' CARA objective:

$$-\mathbb{E}\exp\left(-\frac{\gamma_E}{c}\frac{\Pi_{im}(z)}{E_{im}}\right).$$

Thus c increases effective risk tolerance while leaving the equity-to-deposit scale $e = 0.10$ unchanged. Unlike Section 5.2, this experiment holds expected rates and bank counts fixed: there is no new entry or change in deposit betas. Deposits, investment, wages and employment still adjust to clear markets. All other parameters remain fixed, and none of the capacity responses is targeted.

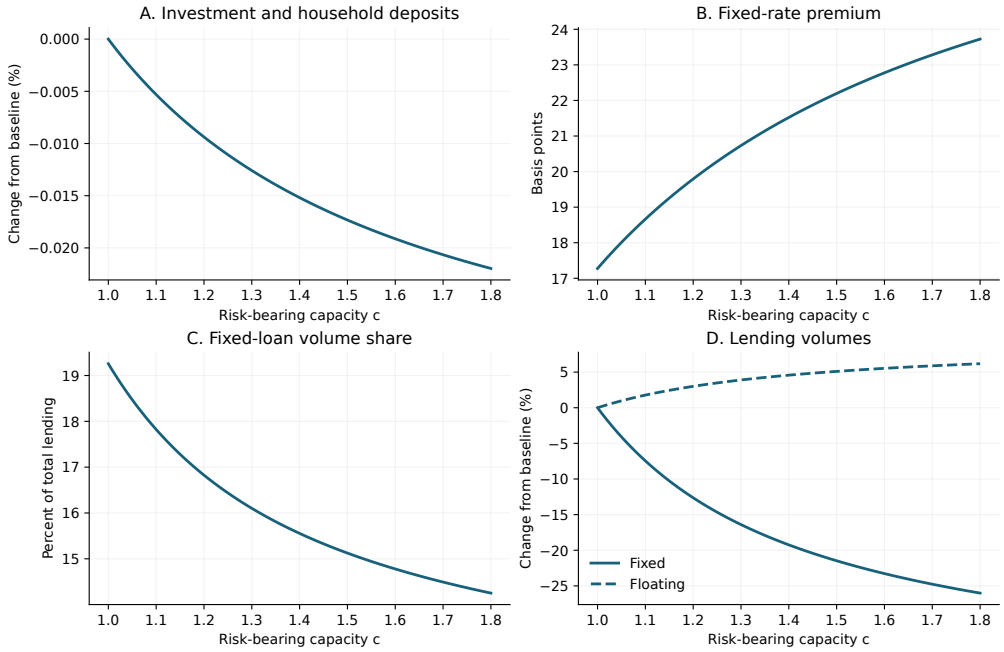


Figure 6: Risk-bearing capacity under the joint CARA calibration

Notes: Capacity changes γ_E/c from the monetary experiment's baseline. Bank counts, expected rates and all other parameters remain fixed; free entry is not reimposed. Panel A reports percentage changes in investment and deposits; Panels B and C report premium and fixed-share levels; Panel D reports lending-volume changes relative to $c = 1$. Premiums use baseline loan-volume weights and fixed shares use current volumes. All responses are untargeted.

Banks remain positively exposed to interest-rate risk along the path in Figure 6. Greater risk tolerance therefore reduces the value of hedging through fixed-rate loans, raising the premium required to supply protection (Panel B). Borrowers switch towards floating finance, reducing the fixed share (Panel C). At $c = 1.8$, the weighted premium rises by 6.45 bp and the fixed share falls by 5.01 pp; fixed lending falls by 26.01% and floating lending rises by 6.18% (Panel D).

Higher fixed borrowing costs also reduce project sizes. Aggregate investment and deposits fall by only 0.02% (Panel A), so the main effect is substitution between contracts. In contrast to the monetary shock, expected floating rates do not rise and entry does not strengthen lending competition. The experiment isolates the loss of banks' hedging incentive rather than the combined effects of higher financing costs and entry.

Diversification or anticipated public support can motivate changes in effective risk tolerance (Chari & Kehoe, 2016; Froot & Stein, 1998). Here c represents that channel, not a structural estimate of either intervention.

6 Conclusion

This paper shows how relative market power in funding and lending shapes banks' interest rate risk and monetary transmission. Banks with less rate-sensitive funding charge higher

fixed–variable loan premiums, even though stable funding makes fixed-rate lending an attractive earnings hedge. Greater lending concentration is associated with higher premiums conditional on funding sensitivity and a stronger expansion of fixed-rate loan stocks after monetary tightening. Our model explains these patterns through banks' tradeoff between earning lending rents and hedging earnings risk. Entry links competition on both sides of the balance sheet, changing both the incentive to hedge and the price of borrower protection.

The joint calibration connects this mechanism to household saving and investment. Following a higher real financing rate, entry lowers aggregate concentration and the fixed-rate premium, while the fixed share of lending rises. Fixed credit expands even as total investment and household deposits contract: cheaper protection partly cushions, but does not offset, higher overall financing costs. The responses differ across markets because stronger lending competition and weaker funding hedges have opposing effects on fixed-rate pricing.

Using the same calibration, greater bank risk-bearing capacity raises premiums and reduces fixed-rate lending when expected rates and bank counts are held fixed. Banks become less willing to supply protection as a hedge, and borrowers switch towards floating finance and reduce project scale. Investment falls as a result. Diversification and anticipated public support may operate through this channel, making lending terms and borrower investment relevant to assessing changes in bank resilience.

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A Data Appendix

A.1 U.S. Small Business Administration 7(a) loans

We use loan-level administrative records from the SBA's 7(a) programme, which supports small businesses unable to obtain conventional credit on reasonable terms. Participating lenders originate loans, while the SBA guarantees a portion of each loan. Proceeds finance working capital, equipment, business premises and acquisitions, as well as refinancing existing debt. Loans may carry fixed or variable interest rates, negotiated between lenders and borrowers subject to programme limits. For variable-rate loans, the repricing frequency is also negotiated between borrower and lender, subject to the requirement that rates adjust no more frequently than monthly. For a detailed overview of the programme, its borrowers and lenders, and loan characteristics, see Gong et al. (2025).

We focus on loans approved during 2010Q1–2019Q4 under the main *Guaranty* subprogramme and issued by FDIC-insured depository institutions. The records identify the lender and report the initial interest rate, rate type, approved and guaranteed amounts, maturity, and borrower location, industry and jobs supported. Lender identifiers allow us to connect loan terms to banks' regulatory reports and branch networks. About one fifth of loans carry fixed rates, both by number and value. Table 5 provides detailed summary statistics.

Table 5: Descriptive statistics of SBA 7(a) loans by rate type, 2010Q1–2019Q4

	Fixed-rate				Variable-rate			
	Mean	P25	P50	P75	Mean	P25	P50	P75
Number of banks	1,013				1,643			
Number of loans	33,207				115,741			
Gross approval ($ billions)	27.8				97.2			
Approval amount ($ thousands)	837	294	560	1,071	840	239	492	1,053
Guarantee share (%)	76.7	75.0	75.0	75.0	77.6	75.0	75.0	75.0
Interest rate at origination (%)	5.86	5.25	5.85	6.50	6.08	5.50	6.00	6.65
Maturity (months)	182	116	120	300	172	120	120	300
Jobs supported	15.8	3.0	7.0	18.0	15.5	3.0	8.0	17.0

Notes. SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks. Approval amount reported in 2019 dollars. The approval amount is the gross approval; the guarantee share is the SBA-guaranteed amount over the gross approval; the rate is the initial contract rate; maturity is the stated term.

A.2 Call Reports and FR Y-9C

We obtain quarterly balance-sheet and income-statement data from the FFIEC 031 Call Reports and the Federal Reserve's Consolidated Financial Statements for Holding Companies (FR Y-9C). Call Reports provide detailed financial information for individual reporting banks, including their assets, liabilities, capital, interest income and interest expense. The FR Y-9C provides corresponding information on a consolidated basis for reporting holding companies. These sources underpin our measures of banks' interest income and expense sensitivities and supply the balance-sheet characteristics used in the loan-pricing analysis.

We consolidate the data at the holding company level using institutional relationship information from the Federal Financial Institutions Examination Council (FFIEC). This unit of observation captures the scope for funding and asset allocation to be coordinated across

affiliated institutions. A holding company can raise funds on behalf of its subsidiaries, while internal capital markets allow resources to be reallocated within the group, influencing subsidiaries' lending decisions (Campello, 2002). Analysing subsidiaries in isolation could therefore omit relevant group-level funding costs and obscure the financial resources supporting their lending. Throughout the paper, "bank" refers to the consolidated entity.

We measure interest income and expense as annualised quarterly flows divided by quarterly average assets. Total interest expense includes payments on deposits, wholesale funding and other liabilities. The resulting expense sensitivity therefore captures the response of overall funding costs per dollar of assets. The regulatory reports also provide the bank size, capital and repricing information used as controls in the empirical analysis.

Following Drechsler et al. (2021), we estimate bank-specific interest expense sensitivities using a distributed-lag specification:

$$\Delta \text{IntExp}_{b,t} = \alpha_b + \eta_t + \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}} \Delta \text{FedFunds}_{t-\tau} + \varepsilon_{b,t}.$$

Here, $\text{IntExp}_{b,t}$ is bank b's annualised interest expense rate: total quarterly interest expense, including payments on deposits, wholesale funding and other liabilities, divided by quarterly average assets and multiplied by four. FedFunds_t denotes the effective federal funds rate, and Δ denotes a quarterly change. The terms α_b and η_t are bank and quarter fixed effects, respectively, and $\varepsilon_{b,t}$ is an error term. The bank-specific coefficients allow funding costs to respond differently across banks, while the contemporaneous rate change and three lags capture adjustment over four quarters.

We summarise each bank's interest expense sensitivity by the sum of these coefficients,

$$\beta_b^{\text{exp}} = \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}}.$$

This measure captures the sensitivity of total funding expense per dollar of assets, rather than the passthrough to deposit rates alone. We construct interest income sensitivity, β_b^{inc}, analogously, replacing the dependent variable with the quarterly change in the annualised interest income rate, defined as four times total quarterly interest income divided by quarterly average assets.

A.3 Summary of Deposits (SOD)

The FDIC Summary of Deposits (SOD) reports annual branch-level information of FDIC-insured institutions as of June 30. It records branch deposit balance, location and institutional identifiers, allowing us to characterise banks' geographic networks and measure local banking market concentration. We treat counties as local markets and branches belonging to the same holding company as a single competitor, consistent with Appendix A.2.

For county c and year t, let $N_{b,c,t}$ denote bank b's number of branches in that county and let $N_{c,t} = \sum_b N_{b,c,t}$ denote the total number of bank branches in county c. We measure county-level concentration using the Herfindahl–Hirschman index (HHI),

$$\text{HHI}_{c,t} = \sum_b \left(\frac{N_{b,c,t}}{N_{c,t}} \right)^2.$$

We then construct a bank-level measure by averaging county HHIs across the bank's branch network, $\mathcal{C}(b)$, weighting each county equally:

$$\text{HHI}_{b,t} = \frac{1}{|\mathcal{C}(b)|} \sum_{c \in \mathcal{C}(b)} \text{HHI}_{c,t},$$

This measure summarises the concentration of banking markets across the bank's geographic network, which we use later as a proxy for bank lending market power, once conditional on bank funding sensitivity β_b^{exp}.

We use $\text{HHI}_{b,t}$ as a proxy for lending-market competition conditional on interest expense sensitivity, β_b^{exp}. Local competition for borrowers can differ from the competitive conditions governing a bank's overall funding costs. Bank holding companies allocate resources across affiliated institutions through internal capital markets, allowing lending in one location to draw on funding raised elsewhere in the group (Campello, 2002). Interest expense sensitivity therefore reflects funding conditions across the bank's geographic network and wholesale markets, whereas competition for a particular borrower depends on locally available lending alternatives.[7] Figure 2 provides supporting evidence for our concentration measure: county-level branch-count HHI is positively and significantly associated with concentration measured using banks' shares of SBA loan dollars. SBA lending shares, however, reflect banks' pricing and credit-supply decisions together with borrower demand, making contemporaneous loan-share HHI potentially endogenous to the fixed–variable rate differential. Using one-year-lagged branch-count HHI mitigates this direct source of simultaneity by measuring established banking presence without incorporating current lending volumes. Its association with SBA lending concentration supports its relevance for the pricing analysis, while conditioning on β_b^{exp} accounts for differences in measured funding-cost sensitivity.

[7]The relevance of locality for lending is particularly strong in the SBA 7(a) market. Di and Pattison (2020) report that 71% of loans in their matched 2017 sample were extended to businesses located within ten miles of a branch of their lender.

B Empirical Appendix

B.1 County-level branch-count concentration

Figure 7 maps branch-count concentration, complementing the SBA loan-share map and the comparison between the two HHIs in Figure 2.

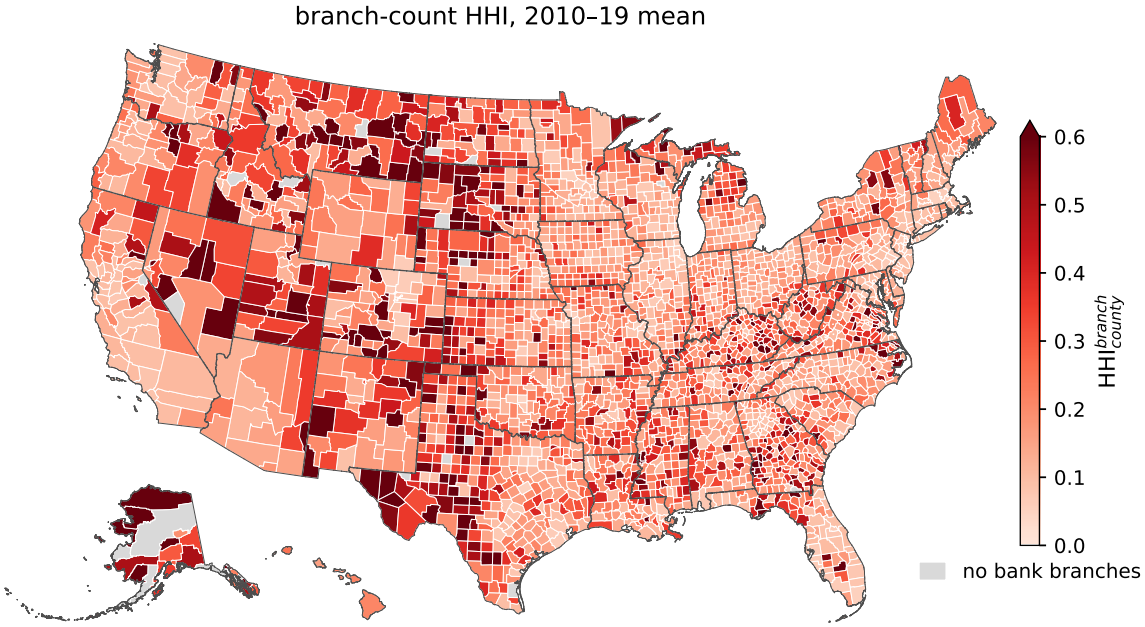


Figure 7: County-level branch-count concentration

Notes: County-level branch-count HHI is the sum of banks' squared shares of local branches. The map reports the mean annual HHI over 2010–2019. Grey counties have no bank branches. Source: FDIC Summary of Deposits.

The distinction between the two measures matters for our pricing analysis. SBA lending shares are equilibrium outcomes that reflect banks' loan pricing and supply decisions together with borrower demand. Concentration calculated from contemporaneous loan shares may therefore be jointly determined with the fixed–variable rate differential. Branch-count concentration captures the distribution of established banking presence without directly incorporating current loan volumes. We accordingly use one-year-lagged branch-count HHI to capture local market structure while reducing this direct source of simultaneity. Its positive association with SBA lending concentration supports its use as a proxy for lending-market concentration, conditional on measured funding sensitivity and the other controls. This interpretation does not require branch networks to be exogenous, but it also does not establish a causal effect of concentration on pricing.

C Model Appendix

C.1 Ownership and the bank objective

Bank owners care about the level and variability of their terminal consumption. We assume that a bank's ownership base is proportional to its assets and that owners share earnings equally. Each owner then bears the earnings risk associated with one unit of lending. This gives the bank objective over returns on assets in Section 3.1 and, when entry is endogenous, the participation value in Section 3.6.

Ownership and preferences. Each bank i has a fixed pool $\overline{\mathcal{O}}_i$ of identical prospective owners, large enough to accommodate the bank sizes under consideration. The pools are disjoint, and an owner can participate only through its designated bank. A bank with assets $d_i > 0$ activates a subset $\mathcal{O}_i \subseteq \overline{\mathcal{O}}_i$ of mass d_i. Asset size and membership are fixed before portfolio choices and the interest-rate shock. Prospective owners outside $\mathcal{O}_i$ remain inactive.

Owners have deterministic outside terminal wealth b and mean–variance preferences over terminal consumption,

$$\mathcal{U}(c) = \mathbb{E}c - \frac{\gamma}{2}\operatorname{Var}(c), \qquad \gamma > 0.$$

The coefficient γ is common across owners and independent of bank size. An inactive owner consumes b, giving value $\mathcal{U}(b) = b$. Owners have no other exposure to the interest-rate shock.

Given bank counts. First hold the number of banks fixed and omit participation costs. Let $\Pi_i(z) = d_i\pi_i(z)$ denote total net interest earnings, where $\pi_i(z)$ is earnings per unit of assets. Equal sharing among the mass d_i of active owners gives each owner terminal consumption

$$c_i^0(z) = b + \frac{\Pi_i(z)}{d_i} = b + \pi_i(z).$$

Since outside wealth is deterministic, the owner's certainty-equivalent gain is

$$\mathcal{U}(c_i^0(z)) - \mathcal{U}(b) = \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \equiv W_i. \tag{30}$$

All owners therefore prefer the fixed-rate portfolio share that solves

$$\max_{l_i^F \in [0,1]} \left\{ \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \right\},$$

taking rival portfolios as given and anticipating the deposit continuation. This is the bank objective in Section 3.1. Mean–variance preferences are specified over owners' consumption, and equal sharing translates them into preferences over the bank's return on assets.

Summing the owners' gains gives

$$d_i W_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)),$$

where $\operatorname{Var}(\Pi_i(z)) = d_i^2 \operatorname{Var}(\pi_i(z))$. Dollar risk tolerance is d_i/γ, proportional to the mass of owners. This ownership interpretation is consistent with Wilson (1968), who shows that a syndicate's risk tolerance equals the sum of its members' risk tolerances under efficient sharing. In the symmetric benchmark, total assets are one and $d_i = 1/N$. Each bank then

has risk tolerance $1/(N\gamma)$, and aggregate risk tolerance across banks is $1/\gamma$. Increasing N divides both assets and active ownership among more banks.

Endogenous entry and participation costs. Now let banks choose whether to operate, with assets assigned according to the active bank counts as in Section 3.6. Ownership follows the same allocation rule, so a participating bank with assets d_i has active owner mass d_i. Participation costs $F > 0$, paid from owners' outside resources in terminal-value units. Active owners share the cost equally, giving each terminal consumption

$$c_i^F(z) = b + \frac{\Pi_i(z) - F}{d_i} = b + \pi_i(z) - \frac{F}{d_i}.$$

The allocation satisfies $\int_{\mathcal{O}_i}(c_i^F(z) - b)\,\mathrm{d}j = \Pi_i(z) - F$. Because the fee is deterministic, the owner's net gain is

$$\mathcal{U}(c_i^F(z)) - \mathcal{U}(b) = W_i - \frac{F}{d_i}.$$

Conditional on participation and bank size, every owner chooses the portfolio that maximises W_i. All active owners also agree on participation, which is weakly preferred when $W_i \geq F/d_i$.

Writing $c_i^F(j, z) = b$ for an inactive owner, the sum of gains across the bank's fixed pool of prospective owners is

$$V_i = \int_{\overline{\mathcal{O}}_i} \left[\mathcal{U}(c_i^F(j, z)) - b\right] \mathrm{d}j = d_i W_i - F.$$

In terms of total earnings, this gives

$$V_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)) - F. \tag{31}$$

A larger bank draws on more owners with the same individual risk tolerance and spreads the fixed cost over a larger ownership base. The common interest-rate shock affects all owners. For a given per-asset earnings distribution, dollar earnings variance grows with the square of bank size, while the coefficient on that variance falls in proportion to size.

A related entry formulation appears in Atkeson et al. (2015), who model banks as coalitions of risk-averse traders and compare per-trader certainty-equivalent gains with a fixed entry cost divided by coalition size. Their coalition sizes are given. Here active ownership follows the asset allocation implied by bank counts, so both size and ownership adjust with entry.

Symmetric entry. Consider a symmetric configuration with market size A, M local deposit communities, and n banks per community. Each bank has assets and active owner mass $d = A/(Mn)$. In the deposit retention continuation, $\beta^d = 1 - 1/n$. Evaluating (4) at the symmetric portfolio $l_i^F = L^F(n; M)$ gives per-asset earnings

$$\pi(z) = \frac{r_0 - r_a}{n} + L^F(n; M)s(n; M) + \left[\frac{1}{n} - L^F(n; M)\right]\sigma z.$$

Since $\mathbb{E}z = 0$ and $\mathrm{Var}(z) = 1$, substitution into (30) yields

$$W(n; M) = \frac{r_0 - r_a}{n} + L^F(n; M)s(n; M) - \frac{\gamma}{2}\sigma^2 \left[\frac{1}{n} - L^F(n; M)\right]^2.$$

This is the per-asset franchise value in (12). The first term is the expected funding advantage, the second is premium income from fixed-rate lending, and the final term is the cost of residual earnings risk borne by each owner.

Using $d = A/(Mn)$ in (31), the participating bank's net value is

$$V(n; A, M) = \frac{A}{Mn}W(n; M) - F = Av(n; M) - F, \qquad v(n; M) = \frac{W(n; M)}{Mn}.$$

The owners' participation condition is therefore equivalent to $Av(n; M) \geq F$. At fixed bank counts, a larger market expands both bank assets and active ownership, increasing total certainty-equivalent earnings relative to the fixed participation cost. Entry changes bank size and membership through the allocation rule. An additional entrant evaluates its value using its post-entry size and the reoptimised deposit and lending continuation, as in (35) and (36). This appendix verifies deposit pricing, establishes the value inequalities used in the entry analysis, and characterises the continuation following entry into a single community.

C.2 Deposit-pricing continuation

Let $\mu = r_0 - r_a > 0$ and $c = 1/n$. Risk-neutral depositors rank offers $r_a + \alpha(r_1(z) - r_a)$ by their expected return $r_a + \alpha\mu$, so larger α is preferred even if $r_1(z) < r_a$ in one state. For integer counts, fix any feasible loan-portfolio profile. We construct a continuation in which both eligible banks offer $\alpha = 1$ on contested accounts; the incumbent chooses its offer on uncontested accounts below. The resulting symmetric pricing conditions also define the continuous extension in n and N used in the main text.

Normalise funding expenses by the bank's committed loan book. As a reference, paying r_a on uncontested accounts and $r_1(z)$ on contested accounts gives exposure $G_i^0 = c - l_i^F$. Against rivals' contested-account offers of one, any unilateral bid profile changes funding expense relative to this reference by $t(r_1(z) - r_a)$ with $t \geq 0$. Losing a contested account replaces funding at the same wholesale rate; attracting a rival's account requires an offer above that rate. Raising an uncontested offer also increases t. The bank therefore chooses the net funding adjustment to maximise

$$-\mu t - \frac{\gamma\sigma^2}{2}(c - l_i^F - t)^2, \qquad t \geq 0.$$

Its unique optimum and resulting average pass-through are

$$t_i^* = \left[c - l_i^F - \frac{\mu}{\gamma\sigma^2}\right]_+, \qquad \beta_i^d = 1 - c + t_i^*. \tag{32}$$

Since $0 \leq t_i^* < c$, this optimum is attained by offering $\alpha = t_i^*/c < 1$ on uncontested accounts and one on contested accounts. All deposits are retained under the tie rule. The optimisation covers all unilateral bid deviations, including bids above one to obtain additional funding as a hedge. Thus these offers are best responses after every portfolio profile. We do not claim

uniqueness of all deposit-bidding equilibria.

Proof of Lemma 1. Hold the other banks' fixed quantities constant. Since $L^F = (1/N)\sum_j l_j^F$, a unilateral increase in l_i^F lowers the market premium at rate $\partial s/\partial l_i^F = -\theta\sigma^2/N$. Anticipating its optimal deposit offer, bank i chooses l_i^F to maximise

$$V_i = c\mu + l_i^F s(L^F, \sigma) - \mu t_i^* - \frac{\gamma\sigma^2}{2}(c - l_i^F - t_i^*)^2.$$

The envelope theorem gives

$$\frac{\partial V_i}{\partial l_i^F} = s - \frac{\theta\sigma^2 l_i^F}{N} + \min\{\gamma\sigma^2(c - l_i^F), \mu\}.$$

This derivative is continuous and strictly decreasing in the bank's own share. Away from the switching point, the second derivative is $-2\theta\sigma^2/N$ where $t_i^* > 0$ and $-\sigma^2(2\theta/N + \gamma)$ where the adjustment constraint is strictly binding at $t_i^* = 0$, establishing global strict concavity. At a symmetric allocation the first-order condition is

$$\theta\sigma^2\left[1 - (1 + 1/N)L^F\right] + \min\{\gamma\sigma^2(c - L^F), \mu\} = 0. \tag{33}$$

Its left-hand side is strictly decreasing, positive at zero and negative at one, so it has one interior root, including for real competition indices $n, N \geq 1$. In the maintained regime $t_i^* = 0$ at this root, and the minimum in (33) equals $\gamma\sigma^2(c - L^F) = \gamma\sigma^2 G$. Substituting $s = \theta\sigma^2(1 - L^F)$ and rearranging gives (5). Strict own concavity ensures that each bank's choice is a best response to the common choice of its rivals. □

To characterise the solution in both deposit-pricing regimes, define

$$L_B(n, N) = \frac{\theta + \gamma/n}{\theta(1 + 1/N) + \gamma}, \qquad L_D(N) = \frac{N}{N + 1}\left(1 + \frac{\mu}{\theta\sigma^2}\right).$$

The joint solution is

$$L^{F*} = \min\{L_B, L_D\}, \qquad s^* = \theta\sigma^2(1 - L^{F*}),$$
$$\beta^{d*} = 1 - \frac{1}{n} + \left[\frac{1}{n} - L^{F*} - \frac{\mu}{\gamma\sigma^2}\right]_+.$$

Writing $G_B = 1/n - L_B$, voluntary additional pass-through is inactive exactly when $\mu \geq \gamma\sigma^2 G_B$. Then equations (5) and (6) apply. The condition below ensures this regime uniformly over real $n \geq 1$. For $n \geq 2$ it is automatic: the left-hand side of (33) is positive at $L^F = c$, so the root has $L^F > c$ and hence $t_i^* = 0$.

C.3 A sufficient condition for competition-driven deposit pass-through

Fix $M \geq 1$ and let $N = Mn$ as n varies over $[1, \infty)$. A sufficient condition for $\beta^d = 1 - 1/n$ at every symmetric equilibrium along this comparison is

$$\mu = r_0 - r_a \geq \frac{\gamma\sigma^2\theta}{\theta(M+1) + \gamma M}. \tag{34}$$

To verify it, set $D = \theta(1 + 1/N) + \gamma$ and write $G_B = c - (\theta + \gamma c)/D$. Holding the other argument fixed gives

$$\frac{\partial G_B}{\partial c} = \frac{\theta(1+1/N)}{D} > 0, \qquad \frac{\partial G_B}{\partial N} = -\frac{\theta(\theta+\gamma c)}{N^2 D^2} < 0.$$

Along $c = 1/n$ and $N = Mn$, increasing n lowers c and raises N, so G_B is strictly decreasing. Its maximum occurs at $n = 1$, where

$$G_B(1, M) = \frac{\theta}{\theta(M+1) + \gamma M}.$$

Condition (34) therefore implies $\mu \geq \gamma\sigma^2 G_B$ for every $n \geq 1$, giving $L^{F*} = L_B$ and zero additional pass-through. It also permits equality at $n = 1$; the envelope condition remains valid at that switching point. This is an equilibrium restriction, not a claim that additional pass-through is never optimal after a deviation in loan portfolios; those deviations are covered by (32).

Equivalently, the condition requires market breadth to satisfy

$$M \geq \underline{M} \equiv \frac{\theta}{\theta + \gamma}\left(\frac{\gamma\sigma^2}{\mu} - 1\right).$$

The negative premium–beta result in Proposition 2 also requires $M < \overline{M} \equiv \theta^2/[\gamma(\theta+\gamma)]$. Thus sufficient joint restrictions are

$$\max\{1, \underline{M}\} \leq M < \overline{M}.$$

For continuous market breadth this interval is nonempty exactly when $\overline{M} > 1$ and $\mu > \gamma^2\sigma^2/(\theta+\gamma)$; when M is a literal number of markets, the interval must contain an integer. A larger M helps keep voluntary deposit hedging inactive but can reverse the premium slope, so a lower bound on M alone does not establish the negative relationship. If $\mu \geq \gamma\sigma^2$, then $\underline{M} \leq 0$ and no additional lower bound beyond $M \geq 1$ is needed.

For the entry extension only, the additional assumption $\mu > \gamma\sigma^2$ makes $t_i^* = 0$ after every feasible portfolio profile, including when a local market has one bank. The same argument applies with heterogeneous counts using $c_i = 1/n_i$ and each bank's own asset normalisation. This supplies the continuation used in the entry derivations below.

C.4 Expected utility and symmetric Cournot lending

Throughout this comparison the deposit contracts are given. For either utility function in Section 3.4, the marginal borrower's premium solves

$$u_\varrho(c-s) = \tfrac{1}{2}[u_\varrho(c+\sigma) + u_\varrho(c-\sigma)], \qquad \varrho = 2\theta(1-L).$$

Greater risk aversion reduces the certainty equivalent of the variable loan, so $s_L < 0$. Under both preferences $s(0) > 0$ and $s(1) = 0$, using the risk-neutral limit at $L = 1$.

Proof of Lemma 2. Fix rival quantities and let l be the bank's own fixed share, so $dL/dl = 1/N$. Write $\mu = r_0 - r_a$. Terminal wealth in the two states is $w(\pm 1) = E + (1-\beta^d)\mu + ls(L) \pm \sigma(1-\beta^d - l)$. A marginal fixed loan pays $s + (l/N)s_L \mp \sigma$ in those states. Weighting these payoffs by marginal utility, dividing by average marginal utility, and then setting $l = L$ gives (9) with

$$H = \sigma \frac{u'(w(-1)) - u'(w(+1))}{u'(w(-1)) + u'(w(+1))}.$$

Substituting the marginal utilities and using $w(\pm 1) = W \pm \sigma G$ gives $H = \sigma \tanh(\gamma\sigma G)$ under CARA and $H = \sigma \tanh[\gamma \operatorname{arctanh}(\sigma G/W)]$ under CRRA. For CRRA we restrict attention to primitives giving positive wealth at every feasible portfolio. □

Proof of Proposition 3. Write $R(L) = s + (L/N)s_L$. For CARA, inverse demand is strictly decreasing and concave. Indeed, for $v(\varrho) = \log\cosh(\varrho\sigma)/\varrho$ and $x = \varrho\sigma$, $\varrho^3 v''(\varrho)$ equals $x^2/\cosh^2 x - 2[x\tanh x - \log\cosh x]$. It vanishes at zero and has derivative $-2x^2\tanh(x)/\cosh^2 x < 0$. The cutoff $\varrho = 2\theta(1-L)$ is affine, which gives the claimed curvature. Thus own premium revenue $ls(L)$ is strictly concave in l, and R is strictly decreasing. For CRRA these two shape restrictions are imposed. State wealth adds an affine term to own revenue, so increasing concave utility gives strict own concavity in both cases.

Now $R(0) > 0 > R(1)$, hence L_C is unique and interior. The hedging term decreases in L (under CRRA use the restriction on G/W) and has the sign of $L_H - L$. At L_C the sign of $R + H$ is the sign of $L_H - L_C$, while at L_H it is the sign of $L_C - L_H$. The unique root is between these shares, or equals their common value. If $L_H = 1$, the root is strictly below one. Own concavity makes this common portfolio a best response for every bank. □

Proof of Proposition 4. Let $F = s + [(1-\beta^d)/M]Ls_L + H$. Implicit differentiation gives $ds/d\beta^d = -s_L F_{\beta^d}/F_L$. Since $s_L < 0$ and $F_L < 0$, the premium falls exactly when $F_{\beta^d} = L(-s_L)/M + H_{\beta^d} > 0$. At fixed L, the CARA hedging term gives $-H_{\beta^d} = \gamma\sigma^2/\cosh^2(\gamma\sigma G)$. For CRRA put $x = \sigma G/W$. Differentiating both exposure and expected equity gives

$$-H_{\beta^d} = \frac{\gamma\sigma^2(W - \mu G)}{(W^2 - \sigma^2 G^2)\cosh^2\big(\gamma \operatorname{arctanh} x\big)}.$$

Here $W - \mu G = E + L(s+\mu) > 0$. At zero bank risk aversion $H = H_{\beta^d} = 0$, while $L_C(-s_L)/M > 0$. Since $R'(L_C) < 0$, an interior equilibrium and the strict inequality persist locally by continuity and the implicit function theorem. This comparison varies bank risk aversion while holding borrower preferences fixed. □

C.5 Uncertainty and the welfare cost of lending power

Proof of Proposition 5. Write $b = 1-\beta^d$, $h = 1/N$, and $v(a) = \log\cosh(a\sigma)/a$, with $v(0) = 0$. Premium payments cancel in the sum of monetary certainty equivalents. The risk cost at a symmetric fixed share L is therefore

$$R(L) = \frac{1}{2\theta}\int_0^{2\theta(1-L)} v(a)\,da + \frac{1}{\gamma}\log\cosh\big(\gamma\sigma(b-L)\big).$$

The first term sums the risk premia of borrowers on variable rates; the second prices bank earnings risk. Inverse demand is $s = v(2\theta(1-L))$. Write $B(L) = -R'(L) = s+\sigma\tanh(\gamma\sigma(b-L))$. Since $s_L < 0$, B is strictly decreasing. It changes sign between zero and one. The efficient share solves $B(L^\star) = 0$, while the Cournot share solves $B(L^F) + hL^F s_L(L^F) = 0$. Hence $B(L^F) > 0$, $L^F < L^\star$, and

$$0 < \Delta(\sigma) = \int_{L^F}^{L^\star} B(u)\,du \le (L^\star - L^F)hL^F[-s_L(L^F)].$$

For small σ, Taylor expansion of $\log\cosh x$ gives the mean–variance risk cost in (11) with opposite sign, up to $O(\sigma^4)$ uniformly in L. The two first-order conditions divided by σ^2 converge to their mean–variance counterparts, each with a unique root. Thus $\Delta(\sigma)/\sigma^2$ converges to the strictly positive mean–variance loss per unit variance.

For large σ, fix any $0 < \varepsilon < 1-b$. On $[b, b+\varepsilon]$ the marginal borrower's coefficient $a = 2\theta(1-L)$ stays bounded away from zero. The identity $\log\cosh x = x-\log 2+\log(1+e^{-2x})$ gives $s = \sigma - (\log 2)/a + o(1)$ and $s_L = -2\theta(\log 2)/a^2 + o(1)$ uniformly there. At $L = b$ the bank hedging term is zero, so both marginal conditions are positive for large σ. At $L = b+\varepsilon$ the hedging term is $-\sigma + o(1)$, making B negative, and the Cournot term is negative as well. Both unique roots therefore converge to b. Since s_L remains bounded near b, the upper bound above implies $\Delta(\sigma) \to 0$. The two roots vary continuously with $\sigma > 0$ by strict monotonicity of their marginal conditions, so the loss is continuous. Its positivity and zero limits at both ends imply an interior global maximum. The argument requires neither a unique peak nor feasibility of the deposit-pricing continuation for arbitrarily large shocks. □

C.6 Franchise value and the participation comparative statics

Maintain the entry extension's additional assumption $\mu \equiv r_0 - r_a > \gamma\sigma^2$. Let $x = 1-\beta^d$, and let $\widehat{W}(\beta^d, N)$ be symmetric per-asset value when beta and lender counts are treated as separate arguments. The portfolio condition implies

$$\widehat{W} = x\mu - \frac{\gamma}{2}\sigma^2x^2 + \sigma^2\left(\frac{\theta}{N}+\frac{\gamma}{2}\right)(L^F)^2.$$

The first two terms are nonnegative because $\mu > \gamma\sigma^2$ and $x \in [0,1]$, while the last is positive. Using $L^F_{\beta^d} = -\gamma/D$ and $L^F_N = \theta L^F/(N^2 D)$ gives

$$\widehat{W}_{\beta^d} = -\mu + \gamma\sigma^2 x - \frac{2\gamma\sigma^2(\theta/N + \gamma/2)L^F}{D} < 0,$$
$$\widehat{W}_N = -\frac{\theta^2\sigma^2(L^F)^2(1-1/N)}{N^2 D} \leq 0.$$

Along $\beta^d = 1 - 1/n$, $N = Mn$,

$$W_n = \frac{\widehat{W}_{\beta^d}}{n^2} + M\widehat{W}_N < 0, \qquad v_n = \frac{W_n}{Mn} - \frac{W}{Mn^2} < 0.$$

The per-asset value remains bounded as $n \to \infty$, so $v \to 0$. For $AW(1;M)/M > F$, continuity and monotonicity yield a unique real $n > 1$ solving (13). Log differentiation yields

$$\frac{d\log n}{d\log A} = \frac{1}{1 - nW_n/W} \in (0,1).$$

Since $\beta^d = 1 - 1/n$ and $N = Mn$, this elasticity implies (14) and completes the proof of Proposition 6. Entry grows less than proportionally with market size, but more local rivals bid for deposits and more lenders compete for borrowers.

C.7 Entry with integer bank counts

A single local market: exact integer entry. For $M = 1$, all banks reoptimise deposit and lending choices when entry raises n to $n+1$. An active integer equilibrium with $n \geq 1$ requires

$$Av(n;1) \geq F \geq Av(n+1;1). \tag{35}$$

Incumbents cover F and another entrant cannot profit. If $Av(1;1) < F$, the market is inactive. At equality either zero or one bank can operate.

Define $A_m = F/v(m;1)$. By Appendix C.6, these thresholds are strictly increasing and unbounded. A count $m \geq 1$ satisfies (35) exactly when $A \in [A_m, A_{m+1}]$. Thus market growth raises the bank count in steps, with two adjacent counts possible at a threshold. Both deposit beta and the number of lenders are nondecreasing in A, and Proposition 2 gives the direction of the premium change across counts.

Several local markets: unilateral entry. For $M > 1$, (13) approximates symmetric participation. An individual entrant changes only its home community's count. Banks then differ in size and beta, so valuing entry requires reoptimising their portfolios.

Let $\mathbf{n} = (n_1, \ldots, n_M)$ be a positive integer vector of local counts, with $N = \sum_{j=1}^M n_j$, and let $v_j(\mathbf{n})$ denote a bank's risk-adjusted continuation value in community j, per unit of aggregate market size. Appendix C.8 derives this continuation. An active integer entry equilibrium satisfies, for each community,

$$Av_j(\mathbf{n}) \geq F, \qquad Av_j(\mathbf{n} + \mathbf{e}_j) \leq F, \tag{36}$$

where $\mathbf{e}_j$ adds one bank in its home community j. Banks compare participation with staying

out ex ante, anticipating portfolio adjustment. After entry the cost is sunk.

From symmetry the incumbent's value per unit of market size is $v(n; M)$. Entry in the first community gives $(n+1, n, \ldots, n)$, whereas $(n+1, \ldots, n+1)$ requires entry everywhere. Appendix C.9 illustrates both premium slopes under unilateral entry.

C.8 Continuation equilibrium with unequal local counts

Take a vector $\mathbf{n} = (n_1, \ldots, n_M)$ with every $n_j \geq 1$ and $N = \sum_{j=1}^{M} n_j$. A bank in community j has assets $d_j = A/(Mn_j)$, global loan-capacity weight $w_j = d_j/A = 1/(Mn_j)$, and deposit beta $\beta_j^d = 1 - 1/n_j$. Banks in that community have identical primitives. Let l_j^F denote their common fixed share. Aggregate fixed lending as a fraction of total credit is $L^F = M^{-1} \sum_j l_j^F$.

Write $h = s/\sigma^2$. The portfolio condition, including boundary choices, gives

$$l_j^F(h) = \min\left\{1, \max\left\{0, \frac{h + \gamma/n_j}{\gamma + \theta/(Mn_j)}\right\}\right\}. \tag{37}$$

Market clearing requires

$$h = \theta\left[1 - \frac{1}{M}\sum_{j=1}^{M} l_j^F(h)\right].$$

The left side minus the right side is continuous and strictly increasing. It is negative at $h = 0$, since $l_j^F(0) < 1$ for every j, and positive at $h = \theta$. Thus there is a unique root in $(0, \theta)$. Equation (37) gives global best responses by own concavity.

Uniqueness also holds without imposing within-community symmetry. Index all banks by i, with capacity weights $w_i > 0$ summing to one and $x_i = 1 - \beta_i^d$. The lending game has the strictly concave weighted potential

$$\mathcal{P}(\mathbf{l}) = \theta \sum_i w_i l_i - \frac{\theta}{2}\left(\sum_i w_i l_i\right)^2 - \frac{\theta}{2}\sum_i w_i^2 l_i^2 + \gamma \sum_i w_i x_i l_i - \frac{\gamma}{2}\sum_i w_i l_i^2.$$

Its derivative divided by w_i equals the derivative of bank i's objective divided by σ^2. The Hessian is $-\theta \mathbf{w}\mathbf{w}' - \mathrm{diag}(\theta w_i^2 + \gamma w_i)$, which is negative definite. Hence the Nash portfolio allocation is unique and symmetric among banks with identical local primitives.

With the solution $h(\mathbf{n})$ and shares $l_j^F(\mathbf{n})$, define

$$v_j(\mathbf{n}) = \frac{1}{Mn_j}\left\{\frac{r_0 - r_a}{n_j} + \sigma^2 l_j^F h - \frac{\gamma}{2}\sigma^2\left(\frac{1}{n_j} - l_j^F\right)^2\right\}. \tag{38}$$

A bank's total continuation value before the fee is Av_j. Recompute (37)–(38) at $\mathbf{n} + \mathbf{e}_j$ to evaluate entry. The positive-count entry conditions (36) follow. For symmetric configurations, all communities face the same entry test. We analyse active configurations rather than characterise all equilibria with inactive communities.

C.9 Numerical illustration of integer entry

Table 6 reports symmetric configurations satisfying participation and no-profitable-entry conditions, using a constant participation cost F. The entrant's continuation is evaluated

at $(n+1, n, \ldots, n)$, with all bank portfolios reoptimised. The examples establish that both premium-slope regimes can occur with unilateral entry.

Breadth M	Size A	Local banks n	Lenders N	Deposit beta	Premium (bp)
1	2	12	12	0.9167	59.15
1	4	18	18	0.9444	51.92
12	2	2	24	0.5000	32.12
12	4	3	36	0.6667	33.50

Table 6: Market size and integer entry. Parameters are $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, $r_0 = 0.04$, $r_a = 0.01$, and $F = 0.001$. Each row satisfies participation and no-profitable-entry conditions after recomputing all loan portfolios following entry into one community. These are illustrative equilibria, not a calibration.

For all four rows in Table 6, evaluating these continuation values gives strictly positive incumbent net values and strictly negative entrant net values. These examples illustrate the entry conditions; they do not establish a general existence or uniqueness theorem for the multi-community entry game.

D Investment Model: Derivations and Checks

The quantitative exercises use one joint equilibrium with exact CARA financial preferences, endogenous project sizes and household saving. This appendix records the demand, pricing and market-clearing conditions underlying both experiments.

D.1 Investment and pricing under CARA

For the derivations, abbreviate hours per unit invested as $\ell(w_1) = [(1-\alpha)\bar{p}A_1/w_1]^{1/\alpha}$, so optimal hours are $h = \ell(w_1)k$. Expected surplus R is defined in (17). For borrower risk aversion t, optimised certainty equivalents are

$$U_F(s) = \max_{k\geq 0}\{(R - s + v_F)k - \kappa k^2/2\}, \qquad U_V(t) = \max_{k\geq 0}\{Rk - \kappa k^2/2 - t^{-1}\log\cosh(t\sigma k)\}.$$

The fixed-contract benefit is nonpecuniary and measured in units of profit. It affects contract and scale choices but is not a cash receipt in the goods accounts. Differentiating the variable-loan objective with respect to k gives $R - \kappa k - \sigma\tanh(t\sigma k)$. Its second derivative is $-\kappa - t\sigma^2\operatorname{sech}^2(t\sigma k) < 0$, so the first-order condition uniquely determines an interior optimum. Together with the fixed-loan condition,

$$k_F = \frac{R - s + v_F}{\kappa}, \qquad \kappa k_V(t) + \sigma\tanh(t\sigma k_V(t)) = R.$$

The optimised floating value decreases with t, so $U_V(t^*) = U_F(s)$ defines the cutoff separating floating from fixed borrowers.

At the indifferent type $t^* = \theta_m^* > 0$, $U_F(s) = \kappa k_F^2/2$. Substituting the variable-loan first-order condition into $U_V(t^*)$ and imposing indifference gives, with $x = t^*\sigma k_V(t^*) > 0$,

$$\frac{\kappa}{2}\left[k_F^2 - k_V(t^*)^2\right] = \frac{x\tanh x - \log\cosh x}{t^*} > 0.$$

The numerator is zero at $x = 0$ and has derivative $x\operatorname{sech}^2 x > 0$. Hence $k_F > k_V(\theta_m^*)$ on the interior branch with positive rate uncertainty.

Write $z = (\log t^* - \mu_m)/\tau$, $f = 1 - \Phi(z)$ and $\mathcal{C}(t,k) = t^{-1}\log\cosh(t\sigma k)$. Along the cutoff curve,

$$k_F(z) = \sqrt{2U_V(t^*)/\kappa}, \qquad s(z) = R + v_F - \kappa k_F(z).$$

The envelope theorem yields

$$s_z = \frac{\tau}{k_F}\{\sigma k_V(t^*)\tanh(t^*\sigma k_V(t^*)) - \mathcal{C}(t^*, k_V(t^*))\} > 0.$$

Per entrepreneur, fixed and floating quantities and their cutoff derivatives are

$$q(z) = fk_F, \qquad q_z = -\phi(z)k_F - fs_z/\kappa,$$
$$v(z) = \int_{-\infty}^{z} k_V(e^{\mu_m + \tau u})\phi(u)\,du, \qquad v_z = k_V(t^*)\phi(z).$$

Thus $Q_m^F = \varphi_m q$, $Q_m^V = \varphi_m v$, $s_{m,Q} = s_z/(\varphi_m q_z)$ and $Q_{m,Q}^V = v_z/q_z$. At the indifferent type $k_V(t^*) < k_F$, so $-1 < Q_{m,Q}^V < 0$: some reduction in fixed credit is a reduction in project

scale, rather than a one-for-one switch to floating finance.

Bank certainty equivalents and pricing. For a symmetric bank let $d_m = \mathcal{D}_m/n_m$, $G_m = 1 - \beta_m^d - Q_m^F/\mathcal{D}_m$ and

$$\bar{\pi}_m = (1 - \beta_m^d)(r_0 - r_a) + \frac{Q_m^F s_m + b_V Q_m^V}{\mathcal{D}_m}.$$

Operating earnings per deposit are $\bar{\pi}_m + \sigma G_m z$. CARA over return on equity with coefficient γ_E and $E_m = e d_m$ therefore gives

$$W_m = \bar{\pi}_m - \frac{1}{g}\log\cosh(g\sigma G_m), \qquad g = \gamma_E/e.$$

Under the restricted-access construction below, a unilateral change in fixed lending has price derivative $J s_{m,Q}$ at symmetry, while the maintained variable-loan allocation gives $\partial q_{im}^V/\partial q_{im}^F = Q_{m,Q}^V$. Differentiating the bank's certainty equivalent yields

$$0 = s_m + J\frac{Q_m^F}{n_m} s_{m,Q} + b_V Q_{m,Q}^V + \sigma\tanh(g\sigma G_m).$$

Rearranging gives the pricing condition in Section 4.2.

Restricted lender access and the continuous approximation. For integer $J \geq 1$ and bank counts divisible by J, divide market m into J identical lending segments indexed by ℓ. Each has borrower mass φ_m/J, the same borrower type distribution, and $n_m/J \geq 1$ banks. Each bank serves one segment; borrowers cannot borrow outside it. Deposit search remains market-wide, so $\beta_m^d = p(n_m; \zeta_m)$ and deposits per bank remain $d_m = \mathcal{D}_m/n_m$. Conditional on loan terms, borrower preferences, technology and investment choices are unchanged.

Let $Q_m^j(s; R)$ denote demand from the full borrower population under contract $j \in \{F, V\}$. Segment demand is $Q_{m\ell}^j(s; R) = Q_m^j(s; R)/J$, so its fixed-loan inverse demand is $s_{m\ell}(Q; R) = s_m(JQ; R)$. At symmetry,

$$Q_{m\ell}^F = Q_m^F/J, \qquad q_{im}^F = \frac{Q_m^F/J}{n_m/J} = \frac{Q_m^F}{n_m}, \qquad \frac{\partial s_{m\ell}}{\partial q_{im}^F} = J s_{m,Q}.$$

The last equality holds with segment rivals' quantities fixed. Thus ordinary Cournot competition generates the markup term $J(Q_m^F/n_m)s_{m,Q}$. For variable lending, the common population scaling cancels:

$$\frac{\partial Q_{m\ell}^V}{\partial Q_{m\ell}^F} = \frac{\partial Q_m^V/\partial s}{\partial Q_m^F/\partial s} = Q_{m,Q}^V.$$

We retain the allocation rule that the bank changing fixed lending absorbs this marginal change in segment variable lending; segmentation alone does not derive that rule. Per-bank lending, deposits and equity are unchanged at symmetry, preserving the service-income and hedging terms. This reproduces the pricing condition above.

The calibration extends these symmetric equations continuously to non-integer J, just as it uses continuous bank counts. Non-integer J measures effective segmentation, with $n_m/J \geq 1$ on the reported equilibrium paths; it is not an exact finite-bank partition. Hold-

ing segmentation fixed as entry adjusts preserves the existing symmetric demand, profit, entry and resource equations, and hence the calibration and reported counterfactual solutions. Entry is symmetric across identical segments with the same market-specific hurdle F_m. Market-wide HHI remains $1/n_m$; within-segment HHI is J/n_m, so the calibration maps observed HHI to the broader market. This derivation establishes the symmetric first-order conditions, not global optimality of bank choices or stability against asymmetric segment entry.

D.2 Household saving and implementation

Household utility is log in current consumption and linear in terminal consumption, as in (21). Deposit relationship weights δ_m are independent of borrower populations. Expected gross deposit return is

$$\mathbb{E}R^d = 1 + \sum_m \delta_m\{(1-\beta_m^d)r_a + \beta_m^d r_0\}.$$

The household conditions are

$$C_0^{-1} = \delta_H \mathbb{E}R^d, \qquad \chi_0 L_0^{\nu_0} = w_0/C_0, \qquad \chi_1 L_1^{\nu_1} = w_1.$$

These conditions are imposed in both the monetary and capacity experiments. The terminal wage is determined before the rate fixing but is not held at its baseline across counterfactuals.

The benchmark is conditional on an exogenous real interbank rate. A government facility can implement it by intermediating matching surplus and deficit bank positions at the same state-contingent floating payoff. Since bank balance sheets and current goods-market clearing imply $\mathcal{D} = I$, the facility's net position and statewise tax requirement are zero. This implementation relies on inherited current retail supply contracts and is not a claim that an arbitrary real rate clears a fully flexible-price economy.

Retail prices and baseline normalisations. The final-good aggregator has elasticity ε across retail varieties. Flexible terminal pricing implies intermediate-good price $\bar{p} = (\varepsilon-1)/\varepsilon$, retail output $Y_1 = A_1 \ell^{1-\alpha} I$, and household-owned retail dividends Y_1/ε. Current retail supply prices are set before the surprise. At the initial allocation, the optimal markup gives $p_0^b = \bar{p}$; after the surprise, $p_0 = w_0/A_0$ adjusts while the inherited retail contract remains in force.

We normalise $I^b = L_0^b = L_1^b = 1$, implying $\ell^b = 1$. Given fitted baseline project surplus R^b, the productivity and preference scales are

$$A_1 = \frac{1 + r_0^b + R^b}{\alpha \bar{p}}, \quad \chi_1 = \frac{1-\alpha}{\alpha}(1 + r_0^b + R^b), \quad A_0 = C_0^b + 1, \quad \chi_0 = \bar{p} A_0 / C_0^b.$$

They are fixed in all counterfactuals. These scale choices do not fix subsequent deposits, investment, wages or employment.

Deposit search and counts. The deposit-search rule is $\beta_m^d = \zeta_m(n_m - 1)/[1+\zeta_m(n_m-1)]$. Consequently observed baseline betas and effective counts $n_m^b = 1/H_m^b$ imply $\zeta_m = \beta_m^{d,b}/[(1-$

$\beta_m^{d,b})(n_m^b - 1)]$. Counts are continuous competition indices. The calibration's positive gap $r_0 - r_a > \sigma$ makes deposits dominate the outside return state by state when $\beta_m^d > 0$.

D.3 Equilibrium and resource accounts

Terminal labour supply and labour demand imply

$$\ell = I^{-\nu_1/(\nu_1+\alpha)}, \qquad 1 + r_0 + R = (1 + r_0^b + R^b)I^{-p}, \qquad p = \frac{\nu_1(1-\alpha)}{\nu_1 + \alpha}.$$

Conditional on counts, the real equilibrium can therefore be expressed as

$$\log \frac{1 + r_0 + R}{1 + r_0^b + R^b} + p \log \left[\sum_m \varphi_m \{q_m + v_m\} \right] = 0. \tag{39}$$

This condition and the three pricing equations determine the project surplus and borrower cutoffs. Free entry adds three equations $\mathcal{D}_m W_m / n_m = F_m$ for the monetary experiment. The capacity experiment instead fixes counts at their baseline and replaces g by g/c. Household deposits always satisfy $\mathcal{D} = I$.

Let $J_2 = \sum_m \varphi_m \mathbb{E} k_m^2$, $Q^V = \sum_m Q_m^V$ and $K_B = \sum_m n_m F_m$. Aggregate bank-owner net payouts after overhead, entrepreneur income and household consumption are

$$\begin{aligned} C_1^B(z) &= \sum_m \mathcal{D}_m \{\bar{\pi}_m + \sigma G_m z\} - K_B, \\ C_1^E(z) &= RI - \frac{\kappa}{2} J_2 - \sum_m s_m Q_m^F - \sigma z Q^V, \\ C_1(z) &= w_1 \ell I + R^d(z) I + Y_1/\varepsilon. \end{aligned}$$

The common bank benefit is accounted for as net ancillary-service output $b_V Q^V$ accruing to bank owners. The borrower convenience benefit is nonpecuniary. Summing the cash budgets yields, in each rate state,

$$C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2 + K_B = Y_1 + b_V Q^V.$$

Financial-owner payouts are net income: fixed outside wealth can cover adverse-state losses and cancels from the resource identity. Current resources satisfy $C_0 + I = A_0 L_0$. Local interbank positions $\mathcal{D}_m - I_m$ sum to zero. Thus changes in deposits are endogenous equilibrium changes in household saving, rather than an exogenous funding restriction.

D.4 Numerical checks

Both equilibria match their financial targets, and entrepreneur shares decrease across markets. Direct certainty-equivalent calculations, higher-order integration of borrower demand and finite-difference demand derivatives reproduce the reported solution. Household optimality, bank balance sheets and statewise goods accounts are satisfied.

Entry is locally stable when counts respond to proportional net entry values. All eigenvalues of the coupled entry Jacobian have negative real parts at both equilibria. Wider count scans give positive net entry values below the equilibrium count and negative values

above it when the other counts are fixed at equilibrium. All 54 tested starting count configurations converge under simultaneous entry adjustment, with deposits, wages and loan quantities re-solved along each path. These are numerical stability checks, not a proof of global uniqueness. The detailed numerical records accompany the calibration code; they are not additional calibration targets.